# Certified global annihilation barriers in an infinite XY lattice

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**Abstract**

A numerical saddle does not establish a globally optimal escape route. We certify the annihilation barrier of a neutral dipole at separation twelve in the exact-cosine XY model on the infinite square lattice. Admissible paths are strongly continuous in the homogeneous energy space and end at any physical uniform phase lift. A finite atlas with rigorous control of infinite-dimensional transverse fibers separates a local minimum from every endpoint. Exact-saddle caps and a validated upper path identify the barrier with an asymmetric index-one saddle height relative to the minimum. In unit-coupling normalization, the barrier lies strictly between 0.00567867 and 0.00570116. Reusable a posteriori tests cover signed-bond coercivity, full dual residuals, and complete boundary coverage. An explicit lattice Green estimate controls critical-state and path tails. Subcritical connections then reuse the global certificate on an infinite class of noncompact initial perturbations with prescribed block tails. We prove termination of barrier enclosures at any requested precision on this class, without claiming a general search implementation. Under summable coupling perturbations of total norm at most 0.002, the minimum and two index-one saddles persist and the global barrier remains positive; the continued saddles are not identified as perturbed minimax realizers. Frozen sources and a fresh execution record document the fixed global certificate. Completeness for arbitrary XY configurations remains open.

**Key words.** XY lattice, annihilation barrier, computer-assisted proof, index-one saddle, a posteriori verification, lattice Green function

**MSC codes.** 82B20, 49J35, 65G20

## 1 Introduction

Discrete phase models support pinned defects, saddle-mediated transitions, and competing escape paths. Finding one path with a low energy maximum gives an upper bound on an activation barrier. Proving that this path is globally optimal requires excluding every less expensive path, including those that leave a proposed reaction surface or create additional defects. On an infinite lattice, even a very accurate finite-box saddle leaves an exterior verification problem. A small residual at the interior vertices does not control the residual created at the interface of an extension.

This paper addresses that distinction for a neutral dipole in the square-lattice XY model with the exact interaction $1 - \cos t$. We work in the homogeneous energy Hilbert space $\mathcal{X}$ defined in section 2. Our central result identifies the annihilation barrier over *all* strongly $\mathcal{X}$-continuous paths to any physical uniform lift. It is a variational assertion about the full infinite lattice, rather than a minimum within a frozen two-coordinate family.

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## 1.1 Results and their scope

For a dipole with charges $+1$ and $-1$ at plaquettes $(0,0)$ and $(12,0)$, the computation certifies a minimum $m_0$ and two reflected asymmetric index-one saddles $s_+$ and $s_-$. Their common energy is denoted by $h$. An open bounded set $\mathcal{D} \subset \mathcal{X}$ contains $m_0$, excludes every annihilation endpoint from its closure, and satisfies $E_0 \geq h$ on its entire boundary. A separately validated path from $m_0$ to an endpoint has maximum $h$. Consequently,

$$B_0(m_0) = h - E_0(m_0), \qquad 0.00567867 < B_0(m_0) < 0.00570116. \tag{1}$$

Here the coupling is one, and the numbers are dimensionless. The minimum and saddle assertions refer to general perturbations in $\mathcal{X}$; no symmetry restriction or positivity of every core bond is imposed. The meaning of "all paths" includes the specified strong topology and the endpoint set in eq. (6).

The geometric argument becomes reusable through theorem 3.7. Its hypotheses are a finite collection of rigorous inequalities, together with analytic formulas controlling the exterior. The conditions cover full dual residuals, signed-bond coercivity, saddle index, exact cutting planes, transverse fibers, all chart faces and seams, and endpoint attachment of the upper path. A certificate that fails one test is inconclusive; it does not prove instability or absence of a lower route. The verification format permits adaptation to another instance without asserting that the search for a successful certificate always terminates.

Three analytic extensions complement this fixed example. First, theorem 5.2 gives an explicit Fourier estimate for the orthogonal projection onto lattice gradients. Combined with the stationary equation, it yields an inverse-radius energy tail. Its constants also control the already validated upper path. Second, theorems 6.2 and 6.4 reuse the fixed communication height for an explicit infinite class of initial fields. The class is generated by compact rational blocks with a geometric norm budget and a prescribed spatial support sequence. It allows noncompact perturbations without imposing symmetry. A subcritical connection to $m_0$ transfers the full path-space certificate, and a constructive energy enclosure terminates for every rational accuracy. This is a consequence of the existing global certificate, rather than a search for all mechanisms. The general refinement procedure is proved; the new code checks its constants and representative finite prefixes. Third, theorem 7.2 treats the infinite-dimensional family of interactions

$$E_b(\theta) = \sum_e (1 + b_e)(1 - \cos D_e\theta), \qquad b \in \ell^1(\mathcal{E}), \quad \|b\|_1 \leq 1/500. \tag{2}$$

The minimum and the two distinct index-one saddles continue, the minimum retains its charges, and

$$|B_b(m_b) - B_0(m_0)| \leq 2\,\|b\|_1 . \tag{3}$$

Thus the perturbed global barrier remains strictly positive. The same coupling class admits a uniform inverse-radius tail for the three continued critical states. We also prove local exponential stability for the specified energy-space gradient flow. We do not claim that a continued saddle necessarily realizes the perturbed minimax, or that a mechanism switch has been established.

## 1.2 Relation to prior work

Certification of critical points of cosine energy landscapes is established. Mehta, Hauenstein, and Wales [15] used algebraic certification for finite XY systems with periodic or antiperiodic boundaries.

More recently, Ayala, Choksi, and Wirth [2] validated finite atomistic configurations and transition-state candidates. These results show why neither an XY critical point certificate nor a validated finite-dimensional saddle should be presented as the new conclusion here.

Metastability and depinning for square-lattice systems with periodic interactions, including XY models, were studied variationally by Alicandro, De Luca, Garroni, and Ponsiglione [1]. They established metastable configurations and derived vortex dynamics through minimizing movements and continuum-limit arguments. Here the lattice spacing and the vortex separation are fixed, and the quantity to be certified is the global communication height from a specified metastable state over all strong energy-space escape paths.

For lattice defects, Hudson and Ortner [13] obtained strongly stable multi-dislocation configurations from an isolated-core stability hypothesis and sufficient separation. We verify a particular dipole equilibrium and a global escape barrier at a fixed separation. Hudson [12] derived asymptotic energy barriers in a finite-domain, piecewise-quadratic dislocation model with a specified single-hop path class. Garroni, Petrache, and Spadaro [11] proved clearing-out results for global minimizers of discrete energies with topological singularities. Global minimization and pinned local metastability ask different questions; a clearing-out theorem does not supply a continuous path with a prescribed ceiling from a given metastable state.

Validated transition paths also have direct precedents. Breden and Kuehn [7] rigorously validated saddle connections in a stochastic transition-path setting. Liu, Chen, and Ortner [14] developed stability theory for minimum energy paths in Hilbert spaces under explicit nondegeneracy hypotheses. Berglund, Medvedev, and Simpson [3] analyzed metastability and relevant transition barriers in the finite one-dimensional nearest-neighbor Kuramoto model, also a cosine system. These works preclude a general claim of priority for a saddle-height formula, for infinite-dimensional path theory, or for rigorous transition geometry. Our lower bound instead comes from a complete separating boundary in the energy space of a two-dimensional infinite lattice. It does not assume that a previously selected path is a minimum energy path.

Several ingredients of the certificate also have established analogues. A posteriori existence arguments based on computable residuals and stability bounds have a substantial history; see Ortner [16] for an inverse-function framework for nonmonotone problems. Signed Laplacian positivity and inertia can be expressed through Schur complements and effective resistances [8]. Cubical sublevel computations and their verification are treated by Dłotko and Wanner [9].

Far-field estimates and boundary approximations were developed by Ehrlacher, Ortner, and Shapeev [10]. Braun, Hudson, and Ortner [4] obtained systematic far-field expansions using continuum correctors and discrete multipoles; higher-order boundary conditions were subsequently developed by Braun, Ortner, Wang, and Zhang [6] and by Wei, Braun, Wang, and Zhang [17]. For the present certificate we require explicit constants in matching bond conventions for the critical states and for an entire upper path.

The contribution is therefore a specific conjunction: the exact cosine interaction on the infinite square lattice; unrestricted strong energy-space escape paths; a matched global lower and upper barrier bound; checkable hypotheses exposing the infinite-dimensional reduction; a terminating reuse procedure on a specified infinite input class; and a quantitative perturbation theorem with effective tails in the same coupling class. We do not infer from this comparison that earlier methods are incapable of adaptation, nor assert universal bibliographic priority.

### 1.3 Verification record and organization

The numerical theorem uses frozen proposals and rigorous verification, with the three source archives indexed in the supplementary record. A fresh execution of the standard supervisor reproduced

every phase it invokes, without using cached validated outputs as mathematical premises. This is a reproduction of the same verification software, supported by analytic review; it is not an independent interval implementation or a proof-assistant formalization. The reproducibility record distinguishes these levels of evidence.

Section 2 fixes the topology and normalization. Section 3 gives the reusable a posteriori theorem, and section 4 records its verified instance. Sections 5 to 7 prove the effective-tail, terminating local-class, and coupling-robustness extensions. Section 8 discusses algorithmic scope and the next certification problem.

## 2 Energy space, lifts, and normalization

Let $V = (\mathbb{Z} + \frac{1}{2})^2$ and orient each nearest-neighbor bond in a positive coordinate direction; $\mathcal{E}$ contains each undirected bond exactly once. Write $D_e u = u(y) - u(x)$ for $e = (x, y)$. Define the closed gradient subspace first:

$$\mathcal{G} = \overline{\{D\varphi : \varphi \in C_c(V;\mathbb{R})\}}^{\ell^2(\mathcal{E})}.$$

The homogeneous energy space and its energy are

$$\langle u, v\rangle_{\mathcal{X}} = \sum_e D_e u D_e v, \qquad \mathcal{X} = \{u : V \to \mathbb{R} : Du \in \mathcal{G}\}/\mathbb{R}, \qquad E_0(u) = \sum_e (1 - \cos D_e u). \tag{4}$$

The quotient identifies vertex representatives differing by a constant. The map $D : \mathcal{X} \to \mathcal{G}$ is an isometric isomorphism: each $g \in \mathcal{G}$ has zero oriented sum around every finite cycle, since finite-cycle sums are continuous on $\ell^2(\mathcal{E})$ and vanish on compact gradients. Integrating $g$ along finite lattice paths therefore reconstructs a vertex representative, unique modulo a constant. Thus $\mathcal{X}$ is a Hilbert space, equivalently the homogeneous completion of compactly supported vertex fields represented modulo constants. Finite-path vertex differences are continuous in its norm. The energy is finite, nonnegative, and at most $\|u\|_{\mathcal{X}}^2/2$.

The physically uniform configurations and admissible barriers are

$$\mathcal{K} = \{u \in \mathcal{X} : D_e u \in 2\pi\mathbb{Z} \text{ for every } e\}, \tag{5}$$

$$B_b(u) = \inf_{\substack{\gamma \in C([0,1];\mathcal{X}) \\ \gamma(0)=u,\ \gamma(1)\in\mathcal{K}}} \max_{t\in[0,1]} \{E_b(\gamma(t)) - E_b(u)\}. \tag{6}$$

Write $B_0$ when $b = 0$. Replacing the endpoint set by the single representative zero can change the stated problem. Each energy is invariant under addition of an element of $\mathcal{K}$.

Every nonzero $k \in \mathcal{K}$ satisfies $\|k\|_{\mathcal{X}} \geq 4\pi$. Indeed, its integer gradient has finite support, and a nonconstant integer field has at least four bonds with nonzero difference on the square lattice. Thus the quotient map $\mathcal{X} \to \mathcal{X}/\mathcal{K}$ is a local isometry and continuous paths lift uniquely after an initial lift is fixed, by subdividing the compact time interval into neighborhoods smaller than half this separation. Consequently eq. (6) also describes paths to the physical vacuum in the phase quotient. Convergence on every fixed finite box is a weaker topology: translating a defect to infinity does not supply a strong energy-space path ending at the vacuum.

We use the Riesz gradient $F(u) = \nabla_{\mathcal{X}} E_0(u)$ and its self-adjoint derivative $H(u)$. The basic differential facts will also specify the constants used in the verification.

**Lemma 2.1.** *The energy is smooth on $\mathcal{X}$, and*

$$\langle F(u), v\rangle_{\mathcal{X}} = \sum_e \sin(D_e u) D_e v, \tag{7}$$

$$\langle H(u)v, w\rangle_{\mathcal{X}} = \sum_e \cos(D_e u) D_e v D_e w. \tag{8}$$

*The bond evaluation functional has norm $1/\sqrt{2}$. In particular,*

$$\|H(u)\| \le 1, \qquad \|H(u) - H(v)\| \le \Lambda \|u - v\|_{\mathcal{X}}, \qquad \Lambda = 1/\sqrt{2}. \tag{9}$$

*For $h \in \mathcal{X}$, the gradient and energy Taylor remainders are bounded by*

$$\begin{aligned} \|F(u+h) - F(u) - H(u)h\| &\le \tfrac{\Lambda}{2} \|h\|_{\mathcal{X}}^2, \\ |E_0(u+h) - E_0(u) - \langle F(u), h\rangle_{\mathcal{X}} - \tfrac{1}{2} \langle H(u)h, h\rangle_{\mathcal{X}} | &\le \tfrac{\Lambda}{6} \|h\|_{\mathcal{X}}^3. \end{aligned} \tag{10}$$

*Proof.* The Fourier isometry sends a lattice gradient to $d(k)\widehat{u}(k)$, with $d_j = e^{ik_j} - 1$ and $|d|^2 = 4 - 2\cos k_1 - 2\cos k_2$. The squared norm of one bond evaluation is the normalized torus integral of $|d_j|^2/|d|^2$. The two integrals agree by coordinate exchange, and their sum is one, so each is $1/2$. The value at $k = 0$ does not affect the integral. This also follows from the projection construction in section 5.

Since $|\sin t| \le |t|$, the first variation defines a bounded functional. Cauchy–Schwarz gives $\|H(u)\| \le 1$, and

$$| \langle (H(u) - H(v))a, c\rangle_{\mathcal{X}} | \le \|D(u-v)\|_\infty \|a\|_{\mathcal{X}} \|c\|_{\mathcal{X}} \le \Lambda \|u - v\|_{\mathcal{X}} \|a\|_{\mathcal{X}} \|c\|_{\mathcal{X}}.$$

The same estimate bounds the trilinear third derivative by $\Lambda$. For every $j \ge 2$, placing two gradient factors in $\ell^2$ and the other $j-2$ factors in $\ell^\infty$ bounds the $j$th derivative by $\Lambda^{j-2} \prod_{i=1}^{j} \|v_i\|_{\mathcal{X}}$; the same coefficient-continuity argument proves smoothness. Differentiation follows by dominated remainder estimates on the bond sums; continuity follows from uniform convergence of the coefficient increments in $\ell^\infty$. Integration along a line gives lemma 2.1. □

At the vacuum, $H(0) = I_{\mathcal{X}}$. A coercivity estimate here is measured against $\|Dv\|_2^2$, not against the vertex $\ell^2$ norm. The local vertex Laplacian has spectrum touching zero on the infinite lattice, so this energy-space gap is not an $\ell^2$ spectral gap.

For a plaquette $p$, choose principal bond increments in $(-\pi, \pi)$ and take their oriented sum divided by $2\pi$. If no increment is at a branch endpoint, this defines an integer charge $q_p$. A uniform margin from those endpoints and the bond evaluation bound certify charge preservation under energy-norm corrections. The dipolar state in section 4 has finite total energy and exactly two nonzero charges. We use this charge statement only after its global branch and exterior checks, not as an assumed constraint on escape paths.

## 2.1 Dissipation metric and dynamical interpretation

The choice of metric specifies how the phase field dissipates energy. In this paper the dissipation potential for an instantaneous velocity $v$ is

$$\mathcal{R}(v) = \frac{1}{2} \sum_e |D_e v|^2 = \frac{1}{2} \|v\|_{\mathcal{X}}^2.$$

Thus the velocity is the unique minimizer of $\mathcal{R}(v) + DE_b(u)[v]$ over $\mathcal{X}$. Its Euler equation is

$$\dot{u} = -\nabla_{\mathcal{X}} E_b(u). \tag{11}$$

Its weak form is

$$\sum_e D_e\dot{u}\, D_e v = -\sum_e (1+b_e)\sin(D_e u) D_e v \quad (v \in \mathcal{X}). \tag{12}$$

Writing $D^*$ for the adjoint from bond fields to vertex distributions, this is $D^*D\dot{u} = -D^*((1+b)\sin Du)$, understood weakly. Along a solution,

$$\frac{\mathrm{d}}{\mathrm{d}t} E_b(u(t)) = -\,\|\dot{u}(t)\|_{\mathcal{X}}^2 = -\sum_e |D_e\dot{u}(t)|^2.$$

The metric therefore measures dissipation through differences of phase velocities across bonds, rather than through independent vertex velocities.

This interpretation has a circuit realization. In a finite array of overdamped resistively shunted Josephson junctions with equal shunt conductances, a bond voltage is proportional to its phase difference velocity. After nondimensionalizing, the sum of the resistive and Josephson bond currents has the form $i_e = D_e\dot{u} + J_e \sin(D_e u)$, where $J_e = 1 + b_e$. Kirchhoff conservation at every free vertex gives $D^*i = 0$ in the absence of injected current and noise, which is the weak equation above when the boundary phases are held fixed. The homogeneous current-conservation equation is given in [18, Eq. (13)]; fluctuating twists require the additional equations specified there and are not included here. The corresponding Joule dissipation is proportional to $\sum_e |D_e\dot{u}|^2$. On the infinite lattice we use the finite-energy weak formulation itself, rather than assuming a thermodynamic limit of the driven or noisy finite-array model. In particular, no bounded inverse of the vertex Laplacian on $\ell^2$ is asserted.

The barrier theorem and the dynamical corollary answer distinct questions. The barrier is determined by the energy and the admissible strong-$\mathcal{X}$ path class in (6); changing the mobility while keeping those two objects fixed does not change this variational quantity. The flow above provides a specified dissipative relaxation model and the local stability assertion in corollary 7.3. An energy barrier alone does not validate a connecting orbit, a thermal escape rate, or the thermodynamic Berezinskii–Kosterlitz–Thouless transition. A different mobility, noise law, or path topology requires its own dynamical analysis.

# 3 A posteriori global-barrier certificates

We give sufficient conditions whose numerical content consists of finite matrix inequalities, scalar interval bounds, and a finite boundary ledger. The conditions apply to the full space $\mathcal{X}$. Their verification does not restrict perturbations to a symmetry class or identify a finite-box Hessian with its infinite-lattice counterpart. Throughout this section, $F = \nabla_{\mathcal{X}} E_0$, $H_\theta = DF(\theta)$, and $\Lambda = 1/\sqrt{2}$ is the Hessian Lipschitz constant from lemma 2.1. The weaker bound $\Lambda = 1$ may instead be used in an individual numerical estimate when its sufficient inequalities still hold. In particular, the correction estimates below allow that coarser choice; the displayed cap parameters use the sharp bound, or its rational upper bound 71/100. Unmarked norms and pairings are those of $\mathcal{X}$, with induced operator norms; finite coordinate vectors carry their Euclidean norm.

The input representation matters. Each candidate is a specified analytic finite-energy reference plus a compact dyadic correction. Its exterior energy, force, branch, and edge-increment remainders must have proved bounds. All core–exterior interface bonds are included. Finite arrays alone do not certify a full $\mathcal{X}^*$ residual. The construction below is an a posteriori criterion for such represented data, rather than an algorithm guaranteed to succeed for every configuration.

### 3.1 Full-space coercivity from finite matrices

Let $Q : \mathbb{R}^d \to \mathcal{X}$ have linearly independent compact columns $q_1, \dots, q_d$, and write

$$G = Q^*Q, \qquad P_Q = QG^{-1}Q^*, \qquad Y = \ker Q^*. \tag{13}$$

The case $d = 0$ means $Y = \mathcal{X}$ and no deflation. The projection is exact; an approximate normalization of stored columns is not substituted for $G$.

**Lemma 3.1** (Neumann comparison)**.** *Let $f$ be an algebraically zero-mean vertex load supported in a square of $N \times N$ vertices, $N \geq 2$. Let $L_N$ count only bonds interior to that square. If $f^\sharp$ denotes its Riesz representative in $\mathcal{X}$, then*

$$\left\|f^\sharp\right\|^2 \leq f^T L_N^\dagger f, \qquad \left|f^T L_N^\dagger f - f^T u\right| \leq \|f\|_{\ell^2} \frac{N^2}{4} \|f - L_N u\|_{\ell^2} \tag{14}$$

*for every finite proposal $u$. No mean constraint on $u$ is required.*

*Proof.* In the variational formula $\left\|f^\sharp\right\|^2 = \sup_v (2f[v] - \|v\|^2)$, discard all bonds not interior to the square. The supremum increases to the finite Neumann problem. The constant mode causes no obstruction because $f$ has zero mean. Moreover, $\lambda_1(L_N) = 4\sin^2(\pi/(2N)) \geq 4/N^2$. Choose a solution $u_*$ of $L_N u_* = f$ with the same mean as $u$. Then $\|u_* - u\|_{\ell^2} \leq N^2 \|f - L_N u\|_{\ell^2}/4$, which proves the second inequality by pairing with $f$. □

The square in this lemma must contain the complete load, including both endpoints of every nonzero flux bond used to form its divergence. The inequality is an upper comparison; it gives no lower bound on the infinite dual norm. Mutual load pairings can be enclosed by the same finite solves and residual bounds. A separately proved residual remainder $D^*r$, with $\|r\|_{\ell^2} \leq \varepsilon$, contributes at most $\varepsilon$ to the dual norm.

For the fixed neutral dipole used below, Appendix D proves the finite load's zero mean by compact-flux cancellation and reflection of the analytic reference, without assuming symmetry of the saddle correction. It also derives the explicit exterior dual remainder and includes the core–exterior interface. Thus the mean condition in lemma 3.1 is checked algebraically for these inputs.

**Proposition 3.2** (Deflated signed-resistance test)**.** *Suppose, uniformly over a prescribed set of configurations,*

$$H_\theta[v, v] \geq a_0 \|v\|^2 - \sum_{e \in S} W_e |D_e v|^2, \qquad a_0 > 0, \quad W_e \geq 0, \tag{15}$$

*where $S$ is finite. A containing Neumann square includes both endpoints of every bond in $S$ and the complete supports of $Q$, whose columns vanish on its boundary. Let $R_N = (f_e^T L_N^\dagger f_f)_{e,f \in S}$, where $f_e = \delta_{e^+} - \delta_{e^-}$, and let $B_{ej} = D_e q_j$. If the finite matrix inequality*

$$W^{1/2}(R_N - BG^{-1}B^T)W^{1/2} \preceq a_0 \kappa I, \qquad \kappa < 1, \tag{16}$$

*is verified, then $H_\theta|_Y \succeq \mu I$ on the full infinite complement, where $\mu = a_0(1 - \kappa) > 0$.*

*Proof.* The infinite trace Gramian restricted to $Y$ is $R_\infty^Y = R_\infty - BG^{-1}B^T$. Applying lemma 3.1 to every linear combination of the loads gives $R_\infty \preceq R_N$. Thus the norm squared of the weighted trace map $v \mapsto (\sqrt{W_e} D_e v)_{e \in S}$ on $Y$ is at most $a_0\kappa$. Substitution in (15) proves the claim. The zero boundary values of the reaction columns ensure that their finite and infinite energy products agree. □

The test uses the entire matrix, including mutual resistances. A validated congruence using interval enclosures of all its entries is sufficient. Individual diagonal resistance tests are not equivalent. Finite signed-Laplacian reductions have substantial precedents [8]; the role of proposition 3.2 here is to preserve the direction of the comparison with the infinite complement.

The weights in (15) are obtained from interval enclosures of $\cos(D_e\theta)$, with an analytic positive floor outside $S$. For a chart displacement $w \in Y$, $\|w\| \le R$, the sharper edge estimate

$$|D_e w| \le R\sqrt{\tfrac{1}{2} - b_e^T G^{-1} b_e}, \qquad b_e = (D_e q_1, \dots, D_e q_d)^T, \tag{17}$$

follows by subtracting the projection of the edge evaluation functional. It is used before enclosing the trigonometric weights. Negative core bonds are allowed; positivity is required of the resulting form.

## 3.2 Critical points and exact saddle planes

**Lemma 3.3** (Frozen Newton and Morse index)**.** *Let $\widehat{\theta} \in \mathcal{X}$ and $\left\|F(\widehat{\theta})\right\| \le \delta$. Assume either $H_{\widehat{\theta}} \succeq \gamma I > 0$, or there is a unit $q \in \mathcal{X}$ such that*

$$H_{\widehat{\theta}}[q,q] \le -a < 0, \qquad H_{\widehat{\theta}}|_{q^\perp} \succeq c_\perp I > 0, \qquad \gamma = \min(a, c_\perp). \tag{18}$$

*If*

$$\delta + \tfrac{\Lambda}{2}\rho^2 \le \gamma\rho, \qquad \Lambda\rho < \gamma, \tag{19}$$

*there is a unique critical point $\theta_*$ in $\overline{B}_{\mathcal{X}}(\widehat{\theta}, \rho) = \{u : \left\|u - \widehat{\theta}\right\|_{\mathcal{X}} \le \rho\}$. It is a strict minimum in the first case and has Morse index one in the second. Its Hessian inverse has norm at most $(\gamma - \Lambda\rho)^{-1}$.*

*Proof.* In the saddle case write $v = tq + y$ and set $Jv = -tq + y$. Self-adjointness cancels the cross terms:

$$\left\langle H_{\widehat{\theta}} v, Jv \right\rangle = -t^2 H_{\widehat{\theta}}[q,q] + H_{\widehat{\theta}}[y,y] \ge \gamma \|v\|^2 .$$

Lax–Milgram applied to $JH_{\widehat{\theta}}$ gives an inverse bounded by $1/\gamma$. The positive $q^\perp$ block has a negative scalar Schur complement, so its Morse index is exactly one. No smallness assumption on the off-diagonal block is needed. In the minimum case the inverse bound is immediate. The map $h \mapsto -H_{\widehat{\theta}}^{-1}(F(\widehat{\theta}) + \mathcal{N}(h))$ is a self-map and contraction on the closed radius-$\rho$ ball by $\|\mathcal{N}(h)\| \le \Lambda \|h\|^2 / 2$ and $\mathrm{Lip}(\mathcal{N}) \le \Lambda\rho$. This closed ball is complete. If the first inequality in (19) is strict, the fixed point is also in the open ball. The same diagonal signs persist throughout that ball, proving the index and inverse bounds at the exact critical point. □

The positive blocks in this lemma can be certified by proposition 3.2, with zero or one deflated direction. This is a validation criterion for supplied candidates, not a search or classification of all critical points; compare finite XY certification in [15]. Coercivity of an isolated configuration and superposition arguments also have prior uses in lattice models [13].

Useful energy and charge consequences are

$$|E_0(\theta_*) - E_0(\widehat{\theta})| \le \min\left\{ \frac{\rho^2}{2}, \frac{\delta\rho}{2} + \frac{\Lambda\rho^3}{12} \right\}. \tag{20}$$

For the second bound apply stationary-endpoint quadrature to $g(t) = E_0(\widehat{\theta} + t(\theta_* - \widehat{\theta}))$: $g'(1) = 0$ and $g(1) - g(0) - g'(0)/2 = \int_0^1 (1/2 - t) g''(t)\, dt$. Subtract $g''(1/2)$ in the integral. A global branch

margin greater than $\rho/\sqrt{2}$ preserves all plaquette charges under the correction. At a minimum the correction segment is convex, so its energy never exceeds $E_0(\widehat{\theta})$.

An optimal-height separator needs planes through the exact saddle. For a saddle $s$ validated near $\widehat{s}$, let $q$ be its certified candidate direction, and put

$$\sigma = \|P_Q q\| > 0, \qquad r = P_Q q/\sigma, \qquad \delta_Q = \sqrt{1-\sigma^2} < 1/\sqrt{2}.$$

Choose $A_0 \geq -H_{\widehat{s}}[q,q]$ and $\eta_0 \geq \left\|P_{q^\perp} H_{\widehat{s}} q\right\|$.

**Lemma 3.4** (Projected cap and adjoint offset)**.** *Suppose the candidate complement floor is $c_\perp$ and*

$$\mu = c_\perp(1-\delta_Q^2) - 2\eta_0\delta_Q\sqrt{1-\delta_Q^2} - A_0\delta_Q^2 - \Lambda\rho_s > 0. \tag{21}$$

*Then $H_s|_{r^\perp} \succeq \mu I$. For $0 < R_c < 3\mu/\Lambda$,*

$$E_0(s+h) \geq E_0(s) + k_c\|h\|^2, \quad h \perp r, \quad \|h\| \leq R_c, \qquad k_c = \frac{\mu}{2} - \frac{\Lambda R_c}{6} > 0. \tag{22}$$

*For any adjoint proposal $z$ with $\left\|H_{\widehat{s}} z - r\right\| \leq \eta_z$,*

$$|\langle r, s-\widehat{s}\rangle + \langle F(\widehat{s}), z\rangle| \leq \eta_z\rho_s + \tfrac{1}{2}\|Dz\|_{\ell^\infty}\rho_s^2. \tag{23}$$

*Proof.* For unit $h \perp r$ write $h = tq + y$, where $|t| \leq \delta_Q$ and $\|y\| = \sqrt{1-t^2}$. Its candidate quadratic form is bounded below by $c_\perp(1-t^2) - 2\eta_0|t|\sqrt{1-t^2} - A_0t^2$. This expression is nonincreasing for $0 \leq |t| \leq \delta_Q < 1/\sqrt{2}$. Evaluate it at $\delta_Q$ and subtract $\Lambda\rho_s$ once at operator level. Taylor's theorem at the stationary point $s$ proves (22). To prove (23), test $F(\widehat{s}) + H_{\widehat{s}}(s-\widehat{s}) + \mathcal{N} = 0$ against $z$. The scalar sine remainder gives $|\langle \mathcal{N}, z\rangle| \leq \|Dz\|_{\ell^\infty}\rho_s^2/2$. □

With $a(\theta) = G^{-1}Q^*(\theta - \theta_{\mathrm{ref}})$, the exact plane is

$$\ell(\theta) = \langle r, \theta - s\rangle = n^T a(\theta) - d = 0, \qquad n = Q^* r, \quad d = \langle r, s - \theta_{\mathrm{ref}}\rangle. \tag{24}$$

The adjoint bound encloses $d$ by computable loads and residuals. An interval strip covering this plane is used to test its intersections; the separator itself is cut by the exact plane, not an approximate midpoint plane. A plane through $\widehat{s}$ generally has a nonzero linear energy term and does not establish an exact saddle-height barrier.

## 3.3 Entire fibers and the boundary ledger

Let $A_i \subset \mathbb{R}^d$ be finitely many bounded open coordinate domains, $y_i \in Y$, and $R_i > 0$. Define

$$c_i(a) = \theta_{\mathrm{ref}} + Qa + y_i, \qquad U_i = \{c_i(a) + w : a \in A_i,\ w \in Y,\ \|w\| < R_i\}. \tag{25}$$

Let $A_0$ be a bounded open coordinate disk or polytope and set $D_0 = (\bigcup_i U_i) \cap a^{-1}(A_0)$. For a common exact reference energy $e_{\mathrm{ref}}$, suppose the chart oracles give, on their complete closed parameter cells,

$$H_{c_i(a)+w}|_Y \succeq \mu_i I > 0 \quad (\|w\| \leq R_i), \quad \|P_Y F(c_i(a))\| \leq \beta_i(a), \quad E_0(c_i(a)) - e_{\mathrm{ref}} \geq V_i^-(a). \tag{26}$$

Consequently

$$E_0(c_i(a)+w) - e_{\mathrm{ref}} \geq V_i^-(a) - \beta_i(a)\|w\| + \tfrac{1}{2}\mu_i\|w\|^2. \tag{27}$$

The reference may remain symbolic: relative-energy bounds between compact changes of the same analytic background avoid subtracting broad, unrelated absolute-energy intervals.

**Lemma 3.5** (Fiber minimization and cap lifting)**.** *If $\beta_i(a) < \mu_i R_i$, the fiber has a unique interior minimizer $w_i(a)$ with $\|w_i(a)\| \le \beta_i(a)/\mu_i$. Suppose $a$ is on the exact plane of a saddle satisfying lemma 3.4, and*

$$\|c_i(a) - \widehat{s}\| + \rho_s + \frac{\beta_i(a)}{\mu_i} < R_c. \tag{28}$$

*Then every $w \in Y$, $\|w\| \le R_i$, satisfies*

$$E_0(c_i(a) + w) \ge E_0(s) + k_c \left\|c_i(a) + w_i(a) - s\right\|^2 + \tfrac{1}{2}\mu_i \left\|w - w_i(a)\right\|^2. \tag{29}$$

*Proof.* Strong convexity on the closed Hilbert ball gives weak lower semicontinuity after extending the function by $+\infty$ outside the ball. Weak compactness gives a minimizer. On its boundary the radial derivative is at least $\mu_i R_i^2 - \beta_i(a) R_i > 0$, excluding a boundary minimizer. Strong monotonicity proves uniqueness and the norm bound. Since $r \in \operatorname{ran} Q$, every $w \in Y$ is orthogonal to $r$. Thus the minimizing point remains in the exact saddle plane; (28) puts it inside the cap. Apply (22) there and strong convexity from that point to $w$. □

It is the minimizing point that must fit in the cap, not the entire transverse ball. In verifying (28), retain the orthogonal decomposition

$$\|c_i(a) - \widehat{s}\|^2 = (a - \widehat{a}_s)^T G (a - \widehat{a}_s) + \|y_i - P_Y(\widehat{s} - \theta_{\mathrm{ref}})\|^2.$$

No global identification of minimizers belonging to different charts is needed for these energy inequalities.

Figure 1 separates the local cap and chart-coverage obligations. Because each normal lies in $\operatorname{ran} Q$, its exact cut is constant along the $Y$ fibers. The cap bounds the fiber minimizer; strong convexity then controls all other fiber points. Coverage of a coordinate face instead uses inclusion of its whole possible low-energy fiber ball, as quantified below.

For clarity, we describe the finite boundary tests. Let $h$ be the exact level to be certified and $\overline{h} \ge h - e_{\mathrm{ref}}$ a verified upper enclosure of its relative energy. Subdivide complete closed faces and intersections, including their edges and corners. Each retained cell receives one of the following witnesses.

1. A radial wall of chart $i$ passes if

$$V_i^- - \beta_i R_i + \tfrac{1}{2}\mu_i R_i^2 \ge \overline{h}. \tag{30}$$

2. Any full-fiber piece passes directly if $V_i^- - \beta_i^2/(2\mu_i) \ge \overline{h}$. Otherwise put

$$\Delta_i = \beta_i^2 + 2\mu_i(\overline{h} - V_i^-), \qquad r_i^+ = \frac{\beta_i + \sqrt{\Delta_i}}{\mu_i}$$

when $\Delta_i \ge 0$. A negative $\Delta_i$ excludes the sublevel. Every point with $E_0 < h$ has $\|w\| < r_i^+$. A coordinate-face cell is therefore internal if a chart $j$ contains its complete coordinate cell strictly and

$$\|y_i - y_j\| + \min(R_i, r_i^+) < R_j. \tag{31}$$

The stronger full-ball test replaces the minimum by $R_i$. To prove that a point is internal to $D_0$, the coordinate cell must also lie in $A_0$. Chart overlap alone cannot make a point on the outer coordinate boundary internal to $D_0$.

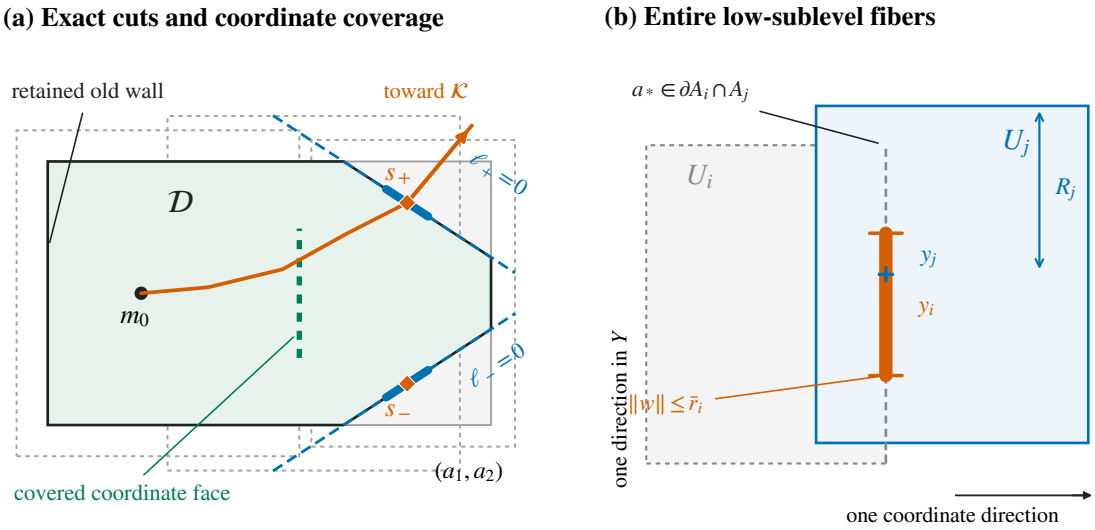



Figure 1: Conceptual certificate geometry; positions and proportions are not numerical atlas data or a computed orbit. (a) Reaction coordinates, with transverse fibers suppressed. The exact cuts pass through $s_\pm$; thick blue segments depict local cap sections. A covered coordinate face is internal, while a retained old wall needs an energy bound. The orange line sketches the upper construction through $s_+$. (b) One coordinate direction and one illustrative direction in $Y$. The entire permitted low-sublevel fiber ball must lie strictly inside the neighboring chart, with the complete coordinate cell inside $A_j$. The displayed Hilbert-norm inequality, rather than overlap in this drawn slice, certifies inclusion in every transverse direction.

3. A new exact-plane face passes by lemma 3.5, uniformly over its intersection with the closed chart, coordinate region, and other cuts. Computations may overcover the plane by the interval strip from (23).

4. Near a cap, a retained outer boundary or seam can instead use the following scalar cone bound. If $H_s[r,r] \ge -A$ and $\|P_{r^\perp} H_s r\| \le \kappa_r$, write $\theta - s = tr + v$, $v \perp r$. Given $|t| \le T$, $\|v\| \in [y_-, y_+]$ and $\|\theta - s\| \le H_0$, set

$$p = \frac{\mu}{2} - \frac{\Lambda H_0}{6} > 0, \quad q_c = \frac{A}{2} + \frac{\Lambda H_0}{6}, \quad y_* = \operatorname{clamp}\left(\frac{\kappa_r T}{2p}, y_-, y_+\right).$$

Then the test

$$p y_*^2 - \kappa_r T y_* - q_c T^2 \ge 0 \tag{32}$$

implies $E_0(\theta) \ge E_0(s)$ throughout the piece. Indeed Taylor's lower bound is $\mu \|v\|^2/2 - \kappa_r |t| \, \|v\| - A t^2/2 - \Lambda(t^2 + \|v\|^2)^{3/2}/6$; replacing the last square root by $H_0$ leaves the displayed quadratic minimum.

All quantities in these tests have interval enclosures obtained from Gram matrices, finite loads, and the specified analytic remainders. A boundary ledger records a complete finite closed cover and the test used on each cell. Pairwise intersections of chart balls, a mesh of boundary samples, or a list omitting tangencies do not establish coverage. In particular the outer coordinate circle is checked as a continuum. The ledger does not assert that a failed test exhibits an actual lower-energy escape.

To make retention by the exact saddle cuts explicit, write

$$H_{\mathrm{cl}} = \bigcap_j \{\ell_j \le 0\}, \qquad H_{\mathrm{op}} = \bigcap_j \{\ell_j < 0\}, \qquad D = D_0 \cap H_{\mathrm{op}}.$$

For each closed old-wall cell $C_\alpha$, the retained set is $C_\alpha \cap H_{\mathrm{cl}}$. The set-level obligation is

$$\theta \in C_\alpha \cap H_{\mathrm{cl}}, \quad E_0(\theta) < h \quad \Longrightarrow \quad \theta \in D_0.$$

An energy bound proves this implication vacuously; a full-fiber overlap witness proves it by strict inclusion in $D_0$. A cell may be discarded by cuts after verifying $C_\alpha \cap H_{\mathrm{cl}} = \varnothing$. Strict positivity of one cut throughout the cell is sufficient but not necessary: a rigorous joint infeasibility test may also establish this. Equality at a cut is retained whenever all other closed inequalities hold. Exclusion from $H_{\mathrm{op}}$ alone does not justify discarding a tangency or an old-wall/new-plane intersection.

Membership and target exclusion are additional finite tests. The entire minimum enclosure must lie strictly inside a chart, the coordinate region, and every retained halfspace. A useful sufficient exclusion is

$$\sup_{\theta \in \overline{D}_0} \|\theta - m\| < 2, \tag{33}$$

when $m$ has a certified charged plaquette. The four principal increments on that plaquette imply $E_0(m) \ge 2$. For every $k \in \mathcal{K}$, $E_0(m) = E_0(m-k) \le \|m-k\|^2/2$, so $\mathrm{dist}_{\mathcal{X}}(m, \mathcal{K}) \ge 2$. The supremum in (33) is bounded using the finite coordinate region, $G$, the chart centers and radii, and the minimum enclosure. Alternatively one may certify charge preservation throughout the closed atlas. Discreteness of $\mathcal{K}$ alone does not locate the atlas relative to all its nonzero lifts.

## 3.4 Upper paths attached to exact critical points

For explicit consecutive nodes $u_j, u_{j+1} \in \mathcal{X}$, the global bound $\|H_\theta\| \le 1$ gives

$$\max_{0 \le t \le 1} E_0((1-t)u_j + t u_{j+1}) \le \max\{E_0(u_j), E_0(u_{j+1})\} + \tfrac{1}{8}\|u_{j+1} - u_j\|^2. \tag{34}$$

This checks the entire time segment. The nodes use coherent real lifts; independent choices of principal phases at successive times do not automatically define an $\mathcal{X}$-continuous path.

**Lemma 3.6** (Stationary-endpoint attachment)**.** *Let $s$ be exact and stationary, $\|s - \widehat{s}\| \le \rho_s$, and $|E_0(s) - E_0(\widehat{s})| \le \varepsilon_s$. For an explicit node $c$, suppose $d_c \ge \|c - \widehat{s}\|$ and $\Delta_c^+ \ge E_0(c) - E_0(\widehat{s})$. If*

$$\Delta_c^+ + \varepsilon_s + \frac{\Lambda}{6}(d_c + \rho_s)^3 < 0, \tag{35}$$

*the segment from $s$ to $c$, except at $s$, has energy strictly below $E_0(s)$.*

*Proof.* For $g(t) = E_0(s + t(c-s))$, one has $g'(0) = 0$ and $|g'''| \le \Lambda(d_c + \rho_s)^3$. Stationary-endpoint Hermite interpolation yields

$$g(t) \le (1-t^2)E_0(s) + t^2 E_0(c) + \frac{\Lambda}{6}(d_c + \rho_s)^3 t^2(1-t).$$

The test makes the difference from $E_0(s)$ strictly negative for $t > 0$. □

Thus no constant energy penalty proportional to $\rho_s^2$ is paid near the saddle. Two attachment nodes enter opposite local descending sides if their $q$-coordinates relative to $\widehat{s}$ have opposite signs and absolute values greater than $\rho_s$. The positive transverse block and negative Schur complement identify the two local strict sublevel components. These are variational attachments, not validated heteroclinic orbits; rigorous transition-path validation has its own antecedents and additional obligations [7].

The minimum is attached to its candidate by the convex bridge from lemma 3.3. Polygonal arms strictly below $E_0(s)$ and two tests (35) then connect that exact minimum through $s$ to a terminal configuration. To reach the physical target, certify zero charge on every terminal plaquette, including the analytic exterior. If $\beta_e \in (-\pi, \pi]$ are its principal increments, then $(D_e\theta - \beta_e)/(2\pi)$ is a curl-free integer bond field of finite support. Integrating it gives a compact integer function $n$, after subtracting its exterior constant. Hence $\theta = 2\pi n + \eta$ with $D\eta = \beta$, and

$$\theta(t) = 2\pi n + (1-t)\eta \tag{36}$$

has nonincreasing energy and ends in $\mathcal{K}$. The finite support follows because a nonzero integer slip requires $|D_e\theta| \geq \pi$, and $D\theta \in \ell^2$. Summability and domination justify continuity and monotonicity of the infinite sum. A finite core check discharges this condition only together with exterior branch and interface checks. The terminal real lift need not be zero; a claim about that particular representative requires the additional verification $n = 0$.

## 3.5 Global closure and reusable conditions

**Theorem 3.7** (A posteriori global-barrier certificate)**.** *Suppose the preceding tests validate a minimum $m$ and finitely many index-one saddles $s_1, \ldots, s_J$. For each saddle choose a cap normal $r_j \in \operatorname{ran} Q$ satisfying lemma 3.4, and write $\ell_j(\theta) = \langle r_j, \theta - s_j \rangle$. Fix a distinguished saddle $s_0$ and put $h = E_0(s_0)$. Assume the finite input certificate verifies the following conditions.*

1. *The exact minimum enclosure is strictly inside $D = D_0 \cap \bigcap_j \{\ell_j < 0\}$, and $\overline{D}_0 \cap \mathcal{K} = \varnothing$.*
2. *A complete closed-cell ledger for every retained old radial, exposed coordinate, and outer-coordinate boundary piece verifies (30), the direct full-fiber bound, (31), or (32) at level $h$. Cells removed by another cut need not pass an energy test; their removal must itself be verified on the complete closed cell relative to $H_{\mathrm{cl}}$. Each retained point of energy below $h$ must be certified strictly internal to $D_0$ by a valid coverage witness. All intersections, edges, corners, and tangencies are included.*
3. *Every new face*
$$\{\theta \in \overline{D}_0 : \ell_j(\theta) = 0,\ \ell_k(\theta) \leq 0\ (k \neq j)\}$$
*is covered by (28) and lemma 3.5, or a direct energy bound at level $h$. For every cap used at its own saddle level, $E_0(s_j) \geq h$ is certified.*
4. *The segment, stationary-endpoint, minimum-bridge, and terminal-lift tests construct a path from $m$ to $\mathcal{K}$ with maximum $h$ through $s_0$.*

*Then, over all continuous strong-$\mathcal{X}$ paths to the physical uniform states,*

$$\inf_{\substack{\gamma \in C([0,1];\mathcal{X}) \\ \gamma(0)=m,\ \gamma(1)\in\mathcal{K}}} \max_t \big(E_0(\gamma(t)) - E_0(m)\big) = h - E_0(m). \tag{37}$$

*Proof.* The earlier tests establish exact critical points, the full-space positive complements, and the exact cap inequalities. For a finite union, $\partial(\bigcup_i U_i) \subset \bigcup_i \partial U_i$. After intersecting with $a^{-1}(A_0)$ and the saddle halfspaces,

$$\partial D \subset (\partial D_0 \cap H_{\mathrm{cl}}) \ \cup \ \bigcup_j (D_0 \cap H_{\mathrm{cl}} \cap \{\ell_j = 0\}).$$

Indeed, every boundary point lies in $\overline{D}_0 \cap H_{\mathrm{cl}}$; if it is in $D_0$ and every cut is strict, it is an interior point of $D$, a contradiction. Thus every boundary point belongs to a retained old wall or a new plane face, including their intersections. On a coordinate wall the overlap test places every hypothetical point below $h$ strictly inside another chart and inside $a^{-1}(A_0)$, so it cannot belong to this boundary. All other ledger witnesses give energy at least $h$. Hence $E_0 \geq h$ throughout $\partial D$. The domain is open, contains $m$, and excludes every endpoint in $\mathcal{K}$. Every admissible path has a first exit and crosses $\partial D$; its maximum is therefore at least $h$. The validated upper path gives the reverse inequality. No compactness of $\partial D$, attainment of the path infimum, or invariant reduced manifold is assumed. □

The energy ordering in the third condition is a real obligation. An exact symmetry preserving the energy and compatible with uniqueness of the validated root can prove equality between reflected saddles. Overlapping numerical energy intervals alone cannot do so.

**Corollary 3.8** (Several competing saddles)**.** *If an upper path is certified through each $s_j$, and every retained old wall is bounded below by $\min_j E_0(s_j)$, then the same construction gives barrier $\min_j E_0(s_j) - E_0(m)$.*

*Proof.* Each cap is bounded below by its own saddle energy and thus by the finite minimum. Choose a saddle attaining that minimum for the upper path. The proof of theorem 3.7 applies without determining in advance which saddle attains the minimum. □

For ledger computations a direct bound by $\min_j \overline{h}_j$ suffices, where $\overline{h}_j \geq E_0(s_j) - e_{\mathrm{ref}}$ are verified relative-energy bounds. A change of dominant mechanism requires an additional certified ordering or crossing argument; uncertainty overlap is not evidence of a crossing. Reuse of the theorem means verifying the same types of obligations on new data. It does not retain old residuals, deficient-bond sets, matrix margins, or boundary coverage automatically when the configuration changes.

# 4 A certified annihilation mechanism at separation twelve

We now instantiate theorem 3.7. All intervals in this section are outward enclosures; the displayed decimal bounds are deliberately rounded away from the certified values. The two candidate states are stored as compact dyadic corrections to the same analytic neutral reference. Every binary floating-point input is interpreted as its exact dyadic value, rather than as an unspecified approximation. Exterior reference estimates and the core–exterior interface terms are included in the verification.

**Theorem 4.1** (Global dipole annihilation barrier)**.** *In the normalization of eq. (4), there exist an exact critical state $m_0 \in \mathcal{X}$ and two distinct exact critical states $s_\pm \in \mathcal{X}$ with the following properties. The state $m_0$ has charge $+1$ at $(0,0)$, charge $-1$ at $(12,0)$, and no other charged plaquettes. It is a strict local minimum. The states $s_\pm$ are asymmetric index-one saddles interchanged by an exact lattice reflection, and $E_0(s_+) = E_0(s_-) = h$. At all three states the Riesz Hessian is invertible with spectral distance from zero greater than $57/1000$.*

*There is a bounded open set $\mathcal{D} \subset \mathcal{X}$ such that*

$$m_0 \in \mathcal{D}, \qquad \overline{\mathcal{D}} \cap \mathcal{K} = \varnothing, \qquad E_0(\theta) \geq h \quad (\theta \in \partial\mathcal{D}). \tag{38}$$

Table 1: Verified inputs and conservative consequences. Candidate quantities are not substituted for exact-state conclusions. The original execution used the valid coarser Hessian Lipschitz constant 1 for the correction deductions shown here.

| Quantity | Bound |
|---|---|
| Minimum candidate coercivity | $> 0.075672$ |
| Minimum correction radius $\rho_m$ | $< 0.01742120$ |
| Exact minimum coercivity | $> 0.05825080$ |
| Saddle candidate negative Rayleigh value | $< -0.06992563$ |
| Saddle candidate complementary coercivity | $> 0.07195499$ |
| Saddle correction radius $\rho_s$ | $< 0.01282494$ |
| Exact saddle inverse-gap lower bound | $> 0.05710069$ |
| Common exact-state gap $\sigma$ | $57/1000$ |
| Exact minimum principal-branch margin | $> 1.1$ |
| Exact cap transverse coercivity (safe value) | $1/25$ |
| Exact cap radius | $33/200$ |
| Reaction-coordinate disk radius | $6599/10000$ |
| Atlas charts / coordinate faces | 109 / 436 |

*There is a continuous path $\Gamma_0$ from $m_0$ to $\mathcal{K}$, passing through $s_+$, with $\max_t E_0(\Gamma_0(t)) = h$. Thus eq. (1) holds for all admissible paths of eq. (6), without restrictions on symmetry, extra defects, or transverse perturbations.*

*Proof.* The frozen data and freshly evaluated inequalities verify the hypotheses of theorem 3.7. The verification has three stages. First, full dual residual bounds and signed-bond coercivity certify the minimum and the index-one saddle near their proposals. Reflection produces the second saddle with exactly equal energy; overlapping numerical energy intervals are not used to infer equality. Second, a 109-chart atlas with two reaction coordinates bounds every retained boundary wall and every exact saddle cutting plane. All transverse directions are retained as Hilbert-space fibers. Third, the validated upper construction attaches its polygonal arms to the exact saddle and to an integer phase lift at the endpoint.

The ledger includes all 436 coordinate faces, radial walls, coordinate-domain boundary portions, and the two cap planes. Five enlarged charts close the remaining plane intersections. The projected-normal and adjoint checks use the exact correction uncertainty, so the planes are not centered at the numerical saddle. The minimum-membership and vacuum-exclusion checks establish eq. (38). The global conclusion now follows from theorem 3.7. The rational energy-difference enclosures give the strict bracket in eq. (1). Numerical margins and the executed supervisor are recorded below and in the supplementary record. □

## 4.1 Margins and certificate geometry

The cap parameters in table 1 use a different, sharper Lipschitz bound than the conservative correction deductions. With $\tau = 71/100$ one has $2\tau^2 = 5041/5000 > 1$, hence $\Lambda = 1/\sqrt{2} < \tau$. Consequently the safe cap values give the entirely rational margin

$$k_c \geq \frac{1}{50} - \frac{71}{100}\frac{33}{1200} = \frac{19}{40000} > 0.$$

The cap checker enforces $R_c < 3\sqrt{2}\mu$, consistently with this choice. Using $\Lambda = 1$ in this cap calculation with only $\mu = 1/25$ would not give a positive coefficient; its separate use for Newton

correction bounds does not require that substitution.

The indefinite Hessian test controls its complete complement and the mutual interactions of the deficient bonds. Its negative direction need not be an eigenvector. The complement estimate together with a negative Rayleigh direction gives invertibility and index one by the reflected-coercivity argument in section 3; the exact-state uncertainty is then subtracted. This distinction matters when extending finite signed-Laplacian tests to an infinite exterior.

The atlas uses a rank-two subspace $Q\mathbb{R}^2$, with orthogonal complement $Y$. For each coordinate cell, the lower bound on the entire transverse fiber is evaluated, not just at its center. A face whose lower bound is too low must be covered strictly by another chart, over the whole closed cell and its entire permitted low-energy transverse ball. The final cutting planes pass through $s_{\pm}$. On those planes the exact saddle energy cancels algebraically, permitting the boundary threshold to equal $h$ without requiring an interval comparison with a positive margin at the saddle itself.

The upper path has only two implicit critical vertices, $m_0$ and $s_+$. Every other vertex except the zero endpoint has the analytic reference plus a compact correction. Interior segments use the quadratic interpolation bound. The two attachments to $s_+$ exploit stationarity at the exact endpoint and a cubic remainder, rather than charging a constant correction error all the way to that endpoint. The terminal contraction is checked after identifying the integer lift on the full lattice. These facts are also the inputs for the path-tail estimate in section 5.

## 4.2 Executed verification and trust boundary

The three frozen archives contain the separator, upper connection, and exact-cap extension. The supplementary index specifies their byte lengths and SHA-256 hashes. They are mathematical inputs with source code, dyadic proposals, and exterior formulas; archived output files are not premises in the fresh run. The command executed in a clean extraction was

```
python3 verify_optimality.py \
  --separator-package ../xy_global_separator \
  --connection-package ../xy_global_connection \
  --output results_fresh --workers 2
```

The final status was `FRESH_OPTIMAL_ANNIHILATION_BARRIER_PASS`. The new run on 21 October 2026 took 737.33 seconds using Python 3.12.14 and g++ 13.3 on x86_64 Linux. The compilation disabled fast arithmetic transformations and floating-point contraction, and enabled rounding-sensitive compilation. The record includes the precise flags, the final certificate hash, all 548 fresh outputs and their hashes, and the integrity check of all 1,808 original archive members. The preceding execution and its resource observations are retained as dated history.

The verifier combines exact rational bookkeeping with outward floating-point enclosures and analytic tail estimates. Candidate search, proposal optimization, and basis generation are outside the supervisor; the fixed proposals must pass its checks. The default supervisor does not execute every historical attack script, the separate secondary boundary implementation, or the separate heteroclinic extension. We therefore base theorem 4.1 on the global variational certificate, not on an unreported fresh heteroclinic calculation.

A further audit on 21 October 2026 rebuilt the standalone separator from its unchanged pinned sources: all 109 local charts and all boundary checks passed. Its 330 directly comparable scientific outputs agreed byte for byte with the archived record. A separately coded exact-dyadic Laplacian calculation verified all 715 geometric interval inclusions and three scalar maxima. The later complete execution described above also rebuilt the connection and optimality assembly from unchanged sources.

Separate instrumented copies check finite ordered interval endpoints and the actual decimal exports against exact binary values. These fixed-data diagnostics and independent rational geometry checks are documented in the supplement. They do not repair the original congruence helper's generic failure to reject an injected NaN; no such failure was found in the protected fixed-data runs. Appendix D derives the residual's zero mean and explicit exterior constants.

The supported reproduction target is little-endian IEEE binary64 input with round-to-nearest arithmetic and a `long double` significand of exactly 64 bits. The companion environment preflight checks this target before a new global run. The frozen extension prints its consumed interval endpoints with an additional outward step and 30 decimal digits; the significance of that padding is tied to this arithmetic format. The source guard allowing at least 64 bits is not a validation of all wider formats. In particular, portability to a 113-bit significand requires a separate serialization analysis or a revised exact-output interface.

Reproduction with the same software tests that the distributed inputs and source produce the asserted output. It does not independently validate the underlying interval implementation or formalize the analytic reductions. The proof in section 3, the full frozen sources, and the supplementary command and integrity record expose those dependencies for examination. The new rational checkers for sections 5 and 7 verify their arithmetic implications; they do not rerun this global supervisor or replace the analytic proofs.

# 5 An explicit Green estimate on the certificate's energy scale

The exterior estimate needed by the certificate concerns the energy norm, including at an index-one saddle. Decay of finite-energy critical fields under stability of the homogeneous lattice is already established in [10, Theorem 2.3]; that regularity theorem does not require strong stability of the critical point. Higher-order far-field conditions and their approximation properties have also been developed for crystalline defects [6, 17]. Our purpose here is quantitative: we give an elementary bound with explicit constants, connect it to the same projections used by the certificate, and retain uniformity under the distributed coupling perturbations considered in this article.

Identify integer coordinates with physical vertices by $x \mapsto (6 + x_1 + \frac{1}{2}, x_2 + \frac{1}{2})$, and use canonical bonds $(x, i) = (x, x + e_i)$, where $x \in \mathbb{Z}^2$ and $i \in \{1, 2\}$. The point $x$ is the *anchor* of the bond. Write $D_i\theta(x) = \theta(x + e_i) - \theta(x)$ and $\mathcal{H} = \ell^2(\mathbb{Z}^2 \times \{1, 2\})$. Each unoriented bond is counted once. Let $\mathsf{P} : \mathcal{H} \to D(\mathcal{X})$ be the orthogonal gradient projection. The range is closed because $D : \mathcal{X} \to \mathcal{H}$ is an isometry onto its image.

**Lemma 5.1** (A numerical kernel bound)**.** *The convolution kernel of* $\mathsf{P}$ *satisfies*

$$|\mathsf{P}_{ij}(z)| \le \frac{8}{(1 + \|z\|_\infty)^2}, \qquad z \in \mathbb{Z}^2, \quad i, j \in \{1, 2\}. \tag{39}$$

*For each source orientation* $j$ *and each integer* $\rho \ge 1$*,*

$$\left( \sum_{\|z\|_\infty \ge \rho} \sum_{i=1}^{2} |\mathsf{P}_{ij}(z)|^2 \right)^{1/2} \le \frac{16\sqrt{2}}{\rho}. \tag{40}$$

*Moreover, the norm of evaluation on any bond is exactly*

$$|D_e v| \le 2^{-1/2} \|v\|_{\mathcal{X}}, \qquad \|D_e\|_{\mathcal{X}^*} = 2^{-1/2}. \tag{41}$$

*Proof.* For $d_i(k) = \exp(ik_i) - 1$, the Fourier symbol is $d_i\overline{d_j}/(|d_1|^2 + |d_2|^2)$ away from the origin. Integrating one coordinate exactly gives $|\mathsf{P}_{ij}(z)| \le \pi/(2r^2) < 2/r^2$ for $r = \|z\|_\infty \ge 1$, with the shifts of the mixed coefficients included. Appendix A supplies that calculation. Since $r+1 \le 2r$, it proves (39); at $z = 0$ the bounded Fourier multiplier gives $|\mathsf{P}_{ij}(0)| \le 1$. There are $8r$ anchors on the shell of radius $r \ge 1$. Therefore

$$\sum_{\|z\|_\infty \ge \rho, i} |\mathsf{P}_{ij}(z)|^2 \le 1024 \sum_{r \ge \rho} \frac{r}{(1+r)^4} \le 1024 \int_\rho^\infty t^{-3}\, dt = \frac{512}{\rho^2}.$$

This proves (40), including both target orientations. Finally $\mathsf{P}_{11}(0) = \mathsf{P}_{22}(0) = 1/2$: the two diagonal symbols sum to one almost everywhere and are exchanged by coordinate symmetry. The squared norm of a coordinate functional restricted to $D(\mathcal{X})$ is $\mathsf{P}_{ii}(0)$. This proves (41) by translation invariance. □

## 5.1 A stationary source and a dyadic tail inequality

The following statement also permits noncompact perturbations of the couplings. Set

$$E_b(\theta) = \sum_e (1 + b_e)(1 - \cos D_e\theta), \qquad b \in \ell^1(\mathcal{E}),$$

so that $E_0 = E_0$. For an integer anchor radius $R \ge 1$, define

$$T_\theta(R) = \left( \sum_{\|x\|_\infty \ge R} \sum_{i=1}^{2} |D_i\theta(x)|^2 \right)^{1/2}. \tag{42}$$

**Theorem 5.2** (An effective $R^{-1}$ tail)**.** *Let $DE_b(\theta) = 0$ in $\mathcal{X}^*$, $\|\theta\|_{\mathcal{X}} \le M$, and suppose that an integer $R_* \ge 1$ and a number $\eta \ge 0$ satisfy*

$$|D_i\theta(x)| \le \eta \quad \text{whenever } \|x\|_\infty \ge R_*.$$

*Put*

$$q_e = D_e\theta - (1 + b_e)\sin(D_e\theta), \qquad Q = \|q\|_{\ell^1},$$
$$a = \frac{\eta^2}{6} + \|b\|_{\ell^\infty}, \qquad R_n = 2^n R_*. \tag{43}$$

*If $2a < 1$, then*

$$Q \le \frac{M^3}{6} + \|b\|_{\ell^1}, \qquad T_\theta(R_n) \le \frac{A}{R_n} \quad (n \ge 0), \tag{44}$$

*where*

$$A = \max\left\{ R_* T_\theta(R_*), \frac{32\sqrt{2}Q}{1 - 2a} \right\}. \tag{45}$$

*For an arbitrary real radius $R \ge R_*$, the bound $T_\theta(R) \le 2A/R$ holds.*

*Proof.* The global inequality $|t - \sin t| \le |t|^3/6$ and $\ell^2 \subset \ell^3$ give $\|D\theta - \sin(D\theta)\|_{\ell^1} \le M^3/6$. Also $\|b\sin(D\theta)\|_{\ell^1} \le \|b\|_{\ell^1}$. Thus $q \in \ell^1 \cap \mathcal{H}$ and the asserted bound for $Q$ holds. Criticality states that $(1 + b)\sin(D\theta)$ is orthogonal to every gradient. Since $D\theta \in D(\mathcal{X})$,

$$D\theta = \mathsf{P}q. \tag{46}$$

This uses the Riesz formulation in $\mathcal{X}$; it assumes no bounded inverse of the vertex Laplacian on $\ell^2(\mathbb{Z}^2)$.

The role of force balance in dislocation far-field estimates is examined in the recent preprint [5]. Here (46) follows directly from stationarity and the closed gradient projection. If it is rewritten as the vertex equation $D^*D\theta = D^*q$, then $q \in \ell^1$ implies $D^*q \in \ell^1$ and $\sum_x (D^*q)(x) = 0$: absolute convergence permits pairing the two opposite endpoint contributions of every bond. Thus this cancellation is proved from the source representation, rather than inferred from far-field decay.

For an even integer $R \geq 2R_*$, split $q = q_{\mathrm{in}} + q_{\mathrm{out}}$ at the anchor radius $R/2$, assigning anchors of radius $R/2$ to the exterior. For a target anchor of radius at least $R$ and a source anchor of radius less than $R/2$, their separation is at least $R/2$. Minkowski's inequality and (40) yield

$$\left\| \mathbf{1}_{\{\|x\|_\infty \geq R\}} \mathsf{P} q_{\mathrm{in}} \right\|_{\mathcal{H}} \leq \frac{32\sqrt{2}}{R} \|q_{\mathrm{in}}\|_{\ell^1} \leq \frac{32\sqrt{2}Q}{R}.$$

On the exterior source set,

$$|q_e| \leq \frac{|D_e\theta|^3}{6} + |b_e|\,|\sin D_e\theta| \leq a|D_e\theta|.$$

The contraction property of $\mathsf{P}$ therefore gives

$$T_\theta(R) \leq \frac{32\sqrt{2}Q}{R} + aT_\theta(R/2). \tag{47}$$

For $A_n = R_n T_\theta(R_n)$ this is $A_n \leq 32\sqrt{2}Q + 2aA_{n-1}$. The interval $[0, A]$, with $A$ as in (45), is invariant under this scalar inequality. Induction proves (44). More explicitly,

$$A_n \leq (2a)^n A_0 + \frac{32\sqrt{2}Q\,[1-(2a)^n]}{1-2a}.$$

If $R_n \leq R < R_{n+1}$, monotonicity gives $T_\theta(R) \leq A/R_n \leq 2A/R$. □

*Remark* 5.3. No support or moment condition on $b$ is used. The exterior perturbation is $b_e \sin(D_e\theta)$, which is controlled *multiplicatively* by $D_e\theta$. Replacing it by an independent additive source would lose this property and would require additional tail information.

## 5.2 The same compact projections and the upper path

Let $\mathcal{X}_S$ contain the configurations with a representative zero outside the integer vertex box $\Lambda_S = [-S, S]^2$, and let $\Pi_S : \mathcal{X} \to \mathcal{X}_S$ be the energy-orthogonal projection. The distinction between anchor and endpoint radii is needed when these projections are compared with stored exterior bounds.

**Lemma 5.4** (A cutoff on the common energy scale)**.** *For every $\theta \in \mathcal{X}$ and integer $R \geq 1$,*

$$\|(I - \Pi_{2R})\theta\|_{\mathcal{X}} \leq \sqrt{227} \left( \sum_{\min(\|x\|_\infty, \|y\|_\infty) \geq R} |D_{\{x,y\}}\theta|^2 \right)^{1/2} \leq \sqrt{227}\, T_\theta(R). \tag{48}$$

*In particular, under Theorem 5.2, the left side at $R = R_n$ is at most $\sqrt{227}A/R_n < 16A/R_n$.*

*Proof.* On $\mathcal{A}_R = \{x : R \le \|x\|_\infty \le 2R\}$, the discrete Poincaré inequality has constant $(225/4)R^2$ (Appendix B). Let $c$ be the annular mean and let $\zeta_R$ equal one up to radius $R$, zero from radius $2R$ onwards, and interpolate linearly between them. The compact representative $\zeta_R(\theta - c)$ gives an admissible approximation in $\mathcal{X}_{2R}$. Using the exact product identity $D(fg) = \overline{f}\, Dg + \overline{g}\, Df$, where bars denote endpoint averages, and $2|\overline{f}|^2 \le |f_x|^2 + |f_y|^2$, its squared error is bounded by

$$2 \sum_{\min(\|x\|_\infty, \|y\|_\infty) \ge R} |D_e\theta|^2 + \frac{4}{R^2} \sum_{\mathcal{A}_R} |\theta - c|^2 \le 227 \sum_{\min(\|x\|_\infty, \|y\|_\infty) \ge R} |D_e\theta|^2.$$

An edge below this minimum-endpoint radius has zero error gradient: integer neighboring radii differ by at most one and the cutoff is still one at radius $R$. Best approximation proves the first inequality. Every edge in that tail has its canonical anchor of radius at least $R$, which proves the second. □

For the homogeneous certificate, the known upper path is polygonal. Its only implicit vertices are the exact minimum and the selected exact saddle. Each remaining vertex, except the zero endpoint, is $\phi_{12}$ plus a stored compact correction. Here

$$\phi_{12}(x) = \arg(6 + x_1 + \tfrac{1}{2} + \mathrm{i}(x_2 + \tfrac{1}{2})) - \arg(-6 + x_1 + \tfrac{1}{2} + \mathrm{i}(x_2 + \tfrac{1}{2})).$$

The argument branches cancel their far-field jumps outside the segment between the two poles. Once $R \ge 16$ and the projection box contains those supports, their projection errors obey the direct dipolar bound $480/R$, proved in Appendix C. Consequently, if $A$ bounds both critical vertices in Theorem 5.2, linearity and convexity of the norm give

$$\sup_t \|(I - \Pi_{2R_n})\Gamma_0(t)\|_{\mathcal{X}} \le \frac{C_\Gamma}{R_n}, \qquad C_\Gamma = \max\{16A, 480\}. \tag{49}$$

No stationary equation is assigned to a segment between vertices.

For finite-box barriers, restrict the paths in (6) to $\mathcal{X}_S$ and the endpoints to $\mathcal{K} \cap \mathcal{X}_S$, retaining all bonds crossing the vertex-box boundary. Denote the resulting barrier by $B_S(u)$ and write $B_\infty = B_0$, $C_\infty(u) = E_0(u) + B_0(u)$. For a compact origin $u \in \mathcal{X}_{2R_n}$, suppose that its segment to $m_0$ lies below the certified level $h$, and that the separator gives $C_\infty(u) = h$. If $M_\Gamma$ bounds the norm on that segment and on $\Gamma_0$, then

$$0 \le B_{2R_n}(u) - B_\infty(u) \le \frac{M_\Gamma C_\Gamma}{R_n}. \tag{50}$$

Indeed, concatenate the segment and $\Gamma_0$ and apply $\Pi_{2R_n}$. The original upper path ends at zero: its terminal check gives `compact_integer_lift_max`=0. Thus the origin and that endpoint are unchanged. Since $\|\nabla_{\mathcal{X}} E_0(z)\|_{\mathcal{X}} \le \|z\|_{\mathcal{X}}$ and $\Pi_{2R_n}$ is a contraction, the energy change of each projected point is at most $M_\Gamma$ times its projection error. This yields the upper bound; the inclusion of finite-box paths in the infinite path class gives the lower bound. Thus the comparison uses the same strong energy topology and keeps all boundary bonds of the finite box.

## 5.3 Numerical consequences and uniform perturbed tails

The payload inequalities in Appendix C permit $M = M_\Gamma = 6007$ for the certified path and for compact origins satisfying $\|u\|_{\mathcal{X}} \le 6007$, with exterior minimum-endpoint radius 4096, and $\eta = 9/500$. The original upper path uses $m_0$ and $s_+$, so anchor radius $R_* = 4097$ is sufficient. Replacing $32\sqrt{2}$ by 46 and $\sqrt{227}$ by 16 gives the rational constants

$$A \le \frac{1246352073472250000}{749919}, \qquad M_\Gamma C_\Gamma \le \frac{119789390485564892000000}{749919}. \tag{51}$$

Exact rational arithmetic gives the following budgets for the allocation $M_\Gamma C_\Gamma / R_n \le \varepsilon/4$:

| $\varepsilon$ | $n$ | $R_n = 4097\,2^n$ | error upper bound |
|---|---|---|---|
| $10^{-2}$ | 54 | 73804990693347688448 | $< 2.165\,10^{-3}$ |
| $10^{-6}$ | 68 | 1209220967519808527532032 | $< 1.321\,10^{-7}$ |

The actual projection box has radius $2R_n$. These are arithmetic bounds, not newly computed barrier intervals. They replace the thousands-of-digits radii from the weak hole-filling estimate by 20- and 25-digit values of $R_n$, but still cannot be materialized. A full cubical covering would also retain its exponential dependence on the finite dimension. Useful radii require sharper data-dependent bounds for $Q$ and $T_\theta(R_*)$ and the retained low-dimensional reaction reduction.

The same estimate is uniform over the nearby critical families of the robustness theorem. For all three critical states, including the reflected saddle, use the common values

$$\|b\|_{\ell^1} \le \frac{1}{500}, \quad M = 6008, \quad \eta = \frac{41}{1000}, \quad R_* = 4098. \tag{52}$$

Here a displacement of at most $1/32$ in $\mathcal{X}$ changes every bond increment by at most $1/(32\sqrt{2})$, so $0.018 + 1/(32\sqrt{2}) < 0.041$. The extra radial unit is deliberate: the physical reflection $x_1 \mapsto 12 - x_1$ sends integer indices $i \mapsto -i - 1$, and a reoriented horizontal bond anchor to $-i - 2$. An anchor radius of 4098 therefore guarantees reflected minimum endpoint radius 4096. With (52),

$$a \le \frac{13681}{6000000}, \qquad Q \le \frac{54216288128003}{1500}, \qquad A \le \frac{4987898507776276000}{2986319} < 1.7\,10^{12}.$$

In particular, all these exact critical states satisfy, on the common dyadic sequence,

$$T_\theta(R_n) \le \frac{1.7\,10^{12}}{R_n}, \qquad \|(I - \Pi_{2R_n})\theta\|_{\mathcal{X}} \le \frac{2.72\,10^{13}}{R_n}. \tag{53}$$

This last conclusion allows noncompact $\ell^1$ coupling perturbations. It is a uniform mathematical tail statement; an implementation accepting arbitrary infinite inputs still requires an effective representation of their coefficients and the relevant membership bounds.

# 6 A terminating certificate on an explicit infinite input class

The global theorem can be reused without searching for a new separator when an input is connected to $m_0$ below its certified communication height. This section specifies an infinite class with a quantitative tail and a terminating enclosure procedure. It concerns the homogeneous interaction $b = 0$. It neither identifies the continued saddles in a perturbed interaction as minimax realizers nor supplies a complete search for arbitrary configurations.

Define the absolute communication height

$$c(u) = \inf_{\gamma(0)=u,\ \gamma(1)\in\mathcal{K}} \max_{0\le t\le 1} E_0(\gamma(t)) = E_0(u) + B_0(u),$$

with the path class of eq. (6). All norms below are in $\mathcal{X}$.

**Lemma 6.1** (Subcritical connection)**.** *Suppose $c(m) = h$, $E_0(m) < h$, and a continuous path from $m$ to $u$ has maximum energy $a < h$. Then $c(u) = h$.*

*Proof.* Concatenation with an arbitrarily near-optimal path from $m$ gives $c(u) \le \max(a, h) = h$. If $c(u) < h$, a path from $u$ to $\mathcal{K}$ with maximum strictly below $h$ exists by the definition of the infimum. Prepending the given connection would contradict $c(m) = h$. No compactness or attainment assertion for a new minimax is used. □

**Theorem 6.2** (Reuse on a full energy ball)**.** *Let $m_0, s_+, h$ be the exact states and height in theorem 4.1, and put $r = 1/20$. For every $u \in \overline{B}_{\mathcal{X}}(m_0, r)$,*

$$c(u) = h, \qquad B_0(u) = h - E_0(u), \qquad 0.00442867 < B_0(u) < 0.00570116. \tag{54}$$

*These inputs have the same two plaquette charges as $m_0$ and no additional charges. They need not themselves be stationary. For the specified flow (11) with $b = 0$, the ball is invariant and*

$$\|u(t) - m_0\| \le e^{-43t/2000} \|u(0) - m_0\|. \tag{55}$$

*Proof.* Stationarity and $H \preceq I$ give, for $v = u - m_0$ and $0 \le t \le 1$,

$$E_0(m_0 + tv) \le E_0(m_0) + \tfrac{1}{2}t^2 \|v\|^2 \le E_0(m_0) + \frac{1}{800} < h.$$

The last strict inequality uses $h - E_0(m_0) > 0.00567867$. Lemma 6.1 proves the height identity. The exact minimum has coercivity greater than $57/1000$. With $\tau = 71/100 > \Lambda$, throughout the ball

$$H(u) \succeq \left(\frac{57}{1000} - \tau r\right) I = \frac{43}{2000} I. \tag{56}$$

Thus $E_0(u) \ge E_0(m_0)$, proving both barrier bounds since $0.00567867 - 1/800 = 0.00442867$. The global principal-branch margin at $m_0$ exceeds 1.1, while every bond changes by at most $r/\sqrt{2} < 1.1$. Charges are therefore preserved along the segment. The gradient is globally Lipschitz. Pairing the evolution with $u - m_0$ and integrating (56) along a segment proves invariance of the ball and (55). □

## 6.1 Inputs with a tail imposed by their construction

Use the integer vertex boxes $\Lambda_R$ and projections $\Pi_R$ of section 5. Set $R_0 = 4098$, $R_n = 2^n R_0$, and $w_n = r2^{-(n+1)}$. An input program supplies a compact rational field $q_n$ supported in $\Lambda_{R_n}$ for every requested integer $n \ge 0$. Define

$$v_n = \frac{w_n}{\max\{1, \|q_n\|^2\}} q_n, \qquad v = \sum_{n=0}^{\infty} v_n, \qquad u = m_0 + v. \tag{57}$$

The norm squared is an exact rational full-lattice bond sum, including all bonds crossing the support boundary. The program is required to return a finite array for every requested $n$; no decision of totality for arbitrary programs is asserted. Call the resulting class $\mathcal{C}_{\mathrm{loc}}$. It permits arbitrary rational block shapes and signs, without symmetry assumptions.

**Proposition 6.3** (Certified localization)**.** *Every input (57) belongs to the ball in theorem 6.2. Its partial sums $p_N = \sum_{n=0}^{N} v_n$ satisfy*

$$\operatorname{supp} p_N \subset \Lambda_{R_N}, \qquad \|v - p_N\| \le r2^{-(N+1)} = \frac{rR_0}{2R_N}. \tag{58}$$

*In particular, with $C_* = 2.72 \times 10^{13}$,*

$$\|(I - \Pi_{2R_N})u\| \le \frac{C_* + rR_0/2}{R_N}. \tag{59}$$

*Proof.* For $t \geq 0$, $t/\max(1,t^2) \leq 1$. Hence $\|v_n\| \leq w_n$; the geometric series gives $\|v\| \leq r$ and (58). Since $p_N \in \mathcal{X}_{2R_N}$, best approximation gives the same bound for $(I - \Pi_{2R_N})v$. The critical-state bound (53), applied with $b = 0$ on its sequence $R_N = 4098\,2^N$, gives the remaining term in (59). □

The class has infinitely many independently variable spatial components. For example, choose $q_n = a_n \delta_{(R_n,0)}$ with rational $|a_n| \leq 1/2$. Then $\|q_n\|^2 = 4a_n^2 \leq 1$, the normalization denominator is one, and nonzero infinitely many $a_n$ give a noncompact perturbation. The tail bounds are consequences of the input format, rather than conclusions guessed from finite exterior samples. Only the energy ball is proved invariant under the flow; preservation of this particular block representation or its spatial tail rate is not needed here and is not asserted.

## 6.2 Effective critical states and finite energy enclosures

A name for $x \in \mathcal{X}$ returns compact rational $x_\delta$ with $\|x - x_\delta\| \leq \delta$ for every positive rational $\delta$. The analytic reference and its exterior estimates in the frozen data provide such names for the candidate states. Finite energy-orthogonal projections are finite Dirichlet systems; their coefficients and the reference's arctangent values can be enclosed to any rational precision. Their projection tails tend to zero. Compact dyadic corrections preserve this property.

For completeness, the nonlinear correction does not require an infinite inverse as an algorithmic primitive. Write $a_e^\sharp$ for the Riesz representative of bond evaluation. For every compact rational field $p$,

$$\left\|a_e^\sharp - p\right\|^2 = \frac{1}{2} - 2D_e p + \|Dp\|_2^2 . \tag{60}$$

Dirichlet projections of $a_e^\sharp$ on increasing boxes are rational and converge in $\mathcal{X}$, by density. Formula (60) is an exact stopping test for their error. If $\|x - x_\delta\| \leq \delta$, then $\|F(x) - F(x_\delta)\| \leq \delta$. The force at the compact field $x_\delta$ is a finite sum of these Riesz vectors with computable sine coefficients. For compact $z$, $H(x)z$ similarly uses only the finitely many bonds on which $D_e z \neq 0$, with computable cosine coefficients. General effective vectors follow by $\|H(x)\| \leq 1$ and density. Thus $F$ and the Hessian action are effective in the same norm.

Let $A = H(\widehat{\theta})$ for either frozen minimum or saddle candidate. The audited bounds imply $\|A\| \leq 1$ and $\left\|A^{-1}\right\| \leq 1/\gamma$ with $\gamma = 69/1000$. Spectral calculus gives

$$A^{-1} f = \sum_{k=0}^{M} A(I - A^2)^k f + \mathcal{R}_M f, \qquad \|\mathcal{R}_M f\| \leq \frac{(1-\gamma^2)^{M+1}}{\gamma} \|f\| . \tag{61}$$

Indeed, $A$ is self-adjoint and its spectrum has modulus between $\gamma$ and one; summing the scalar geometric identity proves the formula also at an index-one candidate. Its finite polynomials can be computed with the preceding oracles, and the remainder tends to zero.

The frozen Newton map $T(z) = z - A^{-1}F(\widehat{\theta}+z)$ is a contraction on its original certified correction ball. Its self-map property uses the original constants in lemma 3.3; substituting a weaker common constant into that self-map test is unnecessary. Since both correction radii are less than $1/50$, (9) and the displayed common inverse bound give contraction factor less than $1/4$. Exact iterates from zero therefore have distance at most $4^{-j}/50$ from the correction. Computing a finite composition to any prescribed additional error is effective by (61). Thus the exact states $m_0, s_+$ have effective names. This is a constructive proof of computability, not an assertion that the original finite-precision supervisor already implements arbitrary-precision refinement.

**Theorem 6.4** (Termination on $\mathcal{C}_{\mathrm{loc}}$)**.** *Given an input program of* (57) *and a positive rational* $\varepsilon$*, there is a procedure using finitely many input blocks, finite linear systems, rational operations, and rigorously enclosed trigonometric evaluations which returns rational* $L, U$ *such that*

$$L \le B_0(u) \le U, \qquad U - L < \varepsilon. \tag{62}$$

*It uses the fixed global certificate once and does not search for a new minimax saddle. No uniform practical complexity bound is asserted.*

*Proof.* The preceding construction gives compact rational names for $s_+$ and, using (58), for $u = m_0 + v$. Both critical states have norm at most $M_* = 6008$ by the existing common Green data. Thus $\|u\| \le M_* + r$. For an approximation $x_\delta$ to an exact field $x$, Taylor expansion at $x$ and $\|F(x)\| \le \|x\|$ give

$$|E_0(x_\delta) - E_0(x)| \le \|x\|\ \delta + \tfrac{1}{2}\delta^2.$$

Choose

$$\delta = \min\left\{1, \frac{\varepsilon}{8(2M_* + r + 1)}\right\}.$$

Compute the two names to this accuracy and enclose each finite bond energy sum with radius at most $\varepsilon/16$, using rational Taylor remainders for cosine. The error in the difference of the exact and compact energies is at most $(2M_* + r)\delta + \delta^2 \le \varepsilon/8$. Adding the two finite-sum errors gives a rational interval of radius at most $\varepsilon/4$. Its width is at most $\varepsilon/2 < \varepsilon$. By theorem 6.2 it encloses $E_0(s_+) - E_0(u) = B_0(u)$. Each requested accuracy entails finitely many blocks and oracle operations, and each refinement loop has a proven vanishing error. This proves termination. □

The compact-name tolerance can be divided between the name of $m_0$ and the block tail; there is no need to materialize the very large projection radius in (59). The latter is a separate spatial localization guarantee. The supplementary rational checker verifies the constants, representative finite block prefixes, and error budgets. The general critical-name and arbitrary-input enclosure procedure above is proved, but is not advertised as an executed general software interface. The substantive global certificate remains theorem 4.1; subcritical concatenation is its elementary reuse principle, not an independent claim of a new universal method.

# 7 Robustness under spatially inhomogeneous couplings

We now extend the single homogeneous certificate to a class of perturbed energies. The extension preserves a positive *global* annihilation barrier and gives a uniform dynamical stability statement. It does not identify the perturbed minimax with the height of a particular continued saddle.

Write $m_0, s_+, s_-$ for the certified unperturbed minimum and saddles, and put

$$h = E_0(s_+) = E_0(s_-), \qquad B_0 = h - E_0(m_0).$$

We use the certified domain $\mathcal{D}$, whose closure misses $\mathcal{K}$, and the certified upper path $\Gamma_0$ from $m_0$ to $\mathcal{K}$, so that

$$m_0 \in \mathcal{D}, \qquad E_0|_{\partial\mathcal{D}} \ge h, \qquad \max_t E_0(\Gamma_0(t)) = h, \qquad .00567867 < B_0 < .00570116. \tag{63}$$

All operator norms and gradients in this section refer to the Riesz identification associated with the $\mathcal{X}$ inner product.

For $b \in \ell^1(\mathcal{E};\mathbb{R})$ define

$$E_b(u) = \sum_{e\in\mathcal{E}} (1+b_e)(1-\cos D_e u) = E_0(u) + W_b(u). \tag{64}$$

The endpoint set remains $\mathcal{K}$ when $1 + b_e > 0$. Denote by $B_b(a)$ the annihilation barrier for $E_b$ starting from $a$: its paths are all strong $\mathcal{X}$-continuous paths from $a$ to $\mathcal{K}$, with no prescribed charge history, symmetry, or exterior configuration.

**Lemma 7.1** (Uniform differential and oscillation bounds)**.** *Let $p = \|b\|_{\ell^1}$. Then $W_b$ is smooth on $\mathcal{X}$, and*

$$\|\nabla_{\mathcal{X}} W_b(u)\| \le \frac{p}{\sqrt{2}}, \qquad \|H_{W_b}(u)\| \le \frac{p}{2}, \tag{65}$$

$$\|H_b(u)\| \le 1 + \frac{p}{2}, \qquad \|H_b(u) - H_b(v)\| \le \frac{1+p/2}{\sqrt{2}} \|u-v\|_{\mathcal{X}}. \tag{66}$$

*Moreover, the absolutely convergent sums*

$$W_- = 2\sum_{b_e<0} b_e, \qquad W_+ = 2\sum_{b_e>0} b_e \tag{67}$$

*satisfy $W_- \le W_b(u) \le W_+$ for every $u \in \mathcal{X}$ and $W_+ - W_- = 2p$.*

*Proof.* Let $g_e$ be the Riesz representative of $D_e$. The nearest-neighbour resistance identity gives $\|g_e\|_{\mathcal{X}} = 1/\sqrt{2}$. Hence

$$\nabla_{\mathcal{X}} W_b(u) = \sum_e b_e \sin(D_e u) g_e, \qquad H_{W_b}(u) = \sum_e b_e \cos(D_e u) g_e \otimes g_e.$$

The series converge absolutely in $\mathcal{X}$ and in operator norm. Their derivatives converge locally uniformly; for derivatives of positive order this also follows directly by summing $|b_e|\,\|g_e\|_{\mathcal{X}}^j$. This proves smoothness and (65). The unperturbed bounds are $\|H_0(u)\| \le 1$ and $\|H_0(u) - H_0(v)\| \le \|u-v\|_{\mathcal{X}}/\sqrt{2}$. For the perturbation, the Lipschitz bound on the cosine gives

$$\|H_{W_b}(u) - H_{W_b}(v)\| \le \sum_e |b_e|\,\|g_e\|_{\mathcal{X}}^3 \|u-v\|_{\mathcal{X}} = \frac{p}{2\sqrt{2}} \|u-v\|_{\mathcal{X}}.$$

This yields (66). Finally for each bond the sign of $b_e$ gives

$$2\min(b_e,0) \le b_e(1-\cos D_e u) \le 2\max(b_e,0).$$

Absolute summability permits adding these inequalities over all bonds. They hold for every compatible phase field, regardless of dependencies between its increments. Thus $W_-$ and $W_+$ are lower and upper enclosures; they need not be attainable extrema, and no independent variation of the bond angles is required. □

The stationary-state certificate provides the following safe rational information:

$$H_0(m_0) \succeq \sigma I, \qquad \left\|H_0(s_\pm)^{-1}\right\| \le \sigma^{-1}, \qquad \operatorname{ind} H_0(s_\pm) = 1, \qquad \sigma = \frac{57}{1000}. \tag{68}$$

Indeed, the minimum's candidate coercivity exceeds .07567 and its correction radius is less than .017422, while the saddle's candidate inverse gap exceeds .069925 and its correction radius is less than .012825. Subtracting the correction radii using the conservative Hessian Lipschitz constant 1 gives .058248 and .057100, respectively, both greater than $\sigma$. The minimum's global principal-increment branch margin is greater than 11/10.

**Theorem 7.2** (A quantitative class of robust global barriers)**.** *Let*

$$\delta = \frac{1}{500}, \qquad r = \frac{1}{32}, \qquad \tau = \frac{71}{100}, \qquad g = \frac{541}{16000}.$$

*For every $b \in \ell^1(\mathcal{E};\mathbb{R})$ with $\|b\|_{\ell^1} \le \delta$, the following assertions hold.*

1. *The energy $E_b$ has unique critical points $m_b, s_b^+, s_b^-$ in the closed radius-$r$ balls about $m_0, s_+, s_-$, respectively. These branches depend smoothly on $b$. The minimum satisfies $H_b(m_b) \succeq gI$; each saddle has index one and inverse norm at most $g^{-1}$. The two saddle branches remain distinct.*

2. *The minimum $m_b$ has precisely the original charges $+1$ and $-1$ at $(0,0)$ and $(12,0)$ and belongs to $\mathcal{D}$.*

3. *Its global annihilation barrier satisfies*

$$\left|B_b(m_b) - B_0\right| \le 2\,\|b\|_{\ell^1}\,. \tag{69}$$

*In particular, throughout this coupling class,*

$$.00167867 < B_b(m_b) < .00970116. \tag{70}$$

*Proof.* Since $\tau^2 > 1/2$, it is a rational upper bound for $1/\sqrt{2}$. Fix an unperturbed critical point $z$ among the three certified states. Taylor expansion gives

$$\nabla_{\mathcal{X}} E_0(z+v) = H_0(z)v + N_z(v), \qquad \|N_z(v)\| \le \frac{\tau}{2}\,\|v\|_{\mathcal{X}}^2\,.$$

On the radius-$r$ ball, $N_z$ has Lipschitz constant at most $\tau r$. The equation $\nabla_{\mathcal{X}} E_b(z+v) = 0$ is equivalent to the fixed point equation

$$v = T_{b,z}(v) := -H_0(z)^{-1}\big[N_z(v) + \nabla_{\mathcal{X}} W_b(z+v)\big].$$

By (68) and Lemma 7.1, the exact self-mapping and contraction tests are

$$\tau\delta + \frac{\tau}{2}r^2 = \frac{45227}{25600000} < \frac{45600}{25600000} = \sigma r, \tag{71}$$

$$\mathrm{Lip}(T_{b,z}) \le \frac{\tau r + \delta/2}{\sigma} = \frac{371}{912} < 1. \tag{72}$$

Banach's theorem produces the unique critical point $z_b$ in this ball. At that point,

$$\|H_b(z_b) - H_0(z)\| \le \tau r + \frac{\delta}{2} = \frac{371}{16000}, \qquad \sigma - \frac{371}{16000} = g. \tag{73}$$

For the minimum this gives the asserted coercivity. For a saddle, (68) implies that the spectrum of its self-adjoint Hessian misses $(-\sigma,\sigma)$. Its negative spectral space is one-dimensional and its complementary spectral space is positive. The perturbation in (73) leaves the quadratic form negative on the former space and positive on the latter. Thus its index is at least one and at most one. The inverse exists by the Neumann-series test, with norm at most $1/(\sigma - 371/16000) = g^{-1}$. This argument does not identify a numerical saddle direction with an exact eigenvector. Smooth dependence on the $\ell^1$ parameter follows from the smooth implicit-function theorem and uniqueness; the strict self-mapping test keeps each root in the interior of its ball.

For completeness, distinctness uses the recorded reaction-coordinate enclosures. If $Q : \mathbb{R}^2 \to \mathcal{X}$ is the reaction embedding and $G = Q^*Q$, the certificate gives $\lambda_{\min}(G) > 1/4$. Therefore the linear functional consisting of the second coordinate of $G^{-1}Q^*u$ has norm less than 2. The candidate saddle coordinates differ by more than .89 in this coordinate, and each original saddle correction has norm less than .013. The exact coordinate difference is consequently greater than $.89 - 4(.013) = .838$, and the original saddle distance is greater than $.419 > .4$. The continued saddle distance is then greater than $.4 - 2r = 27/80$.

Every bond increment changes by at most $\tau r$ between $m_0$ and $m_b$. Its distance from the principal-branch cut remains greater than

$$\frac{11}{10} - \tau r = \frac{3449}{3200} > 0.$$

The straight segment crosses no branch cut, so all plaquette charges are preserved. To establish membership in $\mathcal{D}$, use stationarity of $m_0$ and the global bound $\|H_0\| \le 1$:

$$E_0\big(m_0 + t(m_b - m_0)\big) \le E_0(m_0) + \frac{r^2}{2} = E_0(m_0) + \frac{1}{2048} < h, \qquad 0 \le t \le 1. \tag{74}$$

Since the segment starts in $\mathcal{D}$, it cannot meet its boundary, where (63) gives $E_0 \ge h$. In particular $m_b \in \mathcal{D}$. No inclusion of a full norm ball in $\mathcal{D}$ has been assumed.

It remains to prove the global estimate with the shifted origin. On the radius-$r$ ball about $m_0$, both energies are strictly convex:

$$H_0 \succeq (\sigma - \tau r) I > 0, \qquad H_b \succeq g I > 0.$$

Stationarity at the respective minima and convexity on this ball imply

$$E_0(m_b) \ge E_0(m_0), \qquad E_b(m_b) \le E_b(m_0), \qquad \max_{[m_b, m_0]} E_b \le E_b(m_0). \tag{75}$$

Every admissible path from $m_b$ to $\mathcal{K}$ must cross $\partial\mathcal{D}$. Its maximal perturbed energy is at least $h + W_-$, whereas $E_b(m_b) \le E_b(m_0) \le E_0(m_0) + W_+$. This proves

$$B_b(m_b) \ge B_0 - (W_+ - W_-).$$

For the opposite inequality, prepend the bridge from $m_b$ to $m_0$ to $\Gamma_0$. Its energy is at most $h + W_+$: this follows on $\Gamma_0$ from (63), and on the bridge from (75). Finally $E_b(m_b) \ge E_0(m_0) + W_-$, again by (75). Hence $B_b(m_b) \le B_0 + (W_+ - W_-)$, which proves (69) and (70). □

**Corollary 7.3** (Uniform local stability for the energy-space flow)**.** *For every coupling field in Theorem 7.2, the flow*

$$\dot{u} = -\nabla_{\mathcal{X}} E_b(u) \tag{76}$$

*is globally well posed in $\mathcal{X}$. If $\|u(0) - m_b\|_{\mathcal{X}} \le 1/100$, then the solution stays in that ball and*

$$\|u(t) - m_b\|_{\mathcal{X}} \le e^{-13t/500} \, \|u(0) - m_b\|_{\mathcal{X}}, \qquad t \ge 0. \tag{77}$$

*Along every solution, $\frac{d}{dt} E_b(u) = -\,\|\nabla_{\mathcal{X}} E_b(u)\|_{\mathcal{X}}^2$.*

*Proof.* The gradient is globally Lipschitz, with constant at most $1 + \delta/2 = 1001/1000$, by (66); the Hilbert-space ODE therefore has a unique global solution. Moreover

$$\frac{1 + \delta/2}{\sqrt{2}} < \frac{711}{1000}.$$

Within distance $1/100$ of $m_b$, its Hessian is bounded below by

$$\left(\frac{541}{16000}-\frac{711}{100000}\right) I = \frac{10681}{400000} I \succ \frac{13}{500} I.$$

For $w = u - m_b$, stationarity and integration along the segment from $m_b$ to $u$ give

$$\frac{d}{dt}\frac{\|w\|_{\mathcal{X}}^2}{2} = -\left\langle w, \nabla_{\mathcal{X}} E_b(m_b + w) - \nabla_{\mathcal{X}} E_b(m_b)\right\rangle \le -\frac{13}{500}\|w\|_{\mathcal{X}}^2 .$$

A first-exit argument makes the closed ball invariant; Gronwall's inequality gives (77). The energy identity follows by the chain rule. □

*Remark* 7.4 (Compatibility with the effective exterior estimate). The class above also fits the Green estimate of Section 5, Theorem 5.2. For a continued critical state $z_b$, stationarity gives

$$Dz_b = P_{\mathrm{grad}} q_b, \qquad q_b = Dz_b - (1+b)\sin(Dz_b), \qquad \|q_b\|_{\ell^1} \le \frac{\|z_b\|_{\mathcal{X}}^3}{6} + \|b\|_{\ell^1} .$$

Beyond the certified unperturbed exterior radius, the original increments are bounded by .018. Continuation therefore gives $|D_e z_b| \le .018 + \tau/32 = 643/16000 < .041$. On those exterior bonds,

$$|q_{b,e}| \le \left(\frac{.041^2}{6} + \|b\|_{\ell^\infty}\right)|D_e z_b| \le \frac{13681}{6000000}|D_e z_b|, \qquad 2\frac{13681}{6000000} < 1.$$

Thus the required exterior source term remains absorbable uniformly throughout the $\ell^1$ ball. In particular, a translated coupling perturbation is not an independent additive far-field force: it retains the factor $\sin(D_e z_b)$. No compact-support condition on $b$ is needed for this implication.

*Remark* 7.5 (Scope of the robustness statement). The budget $\|b\|_{\ell^1} \le 1/500$ is a *total* coupling budget. For one modified bond it allows a relative change of 0.2%; it does not allow that change independently on every bond of a growing box. Nonzero constant rescalings of the whole lattice are outside this $\ell^1$ class. The theorem proves neither that one of $s_b^{\pm}$ attains the perturbed minimax nor that reflection breaking selects a different escape mechanism. Those conclusions require additional separator and connection checks. Corollary 7.3 concerns the specified $\mathcal{X}$ mobility, not the local $\ell^2$ gradient flow or a thermal escape rate. The continuation and oscillation arguments are standard; their role here is to turn the reproduced global certificate into an explicit class of robust infinite-lattice conclusions.

# 8 Algorithmic scope and further certification

The a posteriori theorem separates a finite verification task from the search for suitable input data. A successful certificate proves a statement about every admissible escape path in an infinite space. It need not enumerate those paths, classify all critical points, or make the separator invariant under a dynamics. Its reusable content is the collection of explicit inequalities and the requirement that their boundary ledger cover the full continuum.

Theorem 6.4 gives a terminating enclosure procedure on $\mathcal{C}_{\mathrm{loc}}$ by reusing one certified communication height. It does not search for a new saddle or certify arbitrary input states. Its input format supplies effective tails and a subcritical connection. The fixed global chain and the new rational arithmetic checks are executed; the arbitrary-precision critical-name interface used in the constructive theorem has not been implemented as a general program.

A complete algorithm for all effectively represented finite-energy XY configurations remains an open problem. In particular, this paper does not prove that every strict lower level admits a finite separator recognized by its verifier. Such completeness, or a different effective global localization theorem, would be required to justify universal termination. Local spectral reduction alone does not provide it.

This does not give a practical search algorithm for arbitrary XY configurations. An infinite input must come with an effective representation and tail bounds; finite samples cannot certify arbitrary unseen exterior data. Even for compact inputs in the certified local class, a complete cubical search on a large truncation has exponential dimension dependence. The inverse-radius estimate improves the rigorous localization rate substantially, but its current norm majorants still yield impractical truncation sizes. The global barrier in theorem 4.1 was obtained from the tailored atlas and analytic exterior, not from materializing those enormous boxes.

For the perturbation family in eq. (2), the conclusion is stronger than persistence of an isolated finite-box root: it includes the global barrier estimate and tails in the same infinite-dimensional class. It is also weaker than a parameter-uniform identification of a transition saddle. The continued saddles are genuine index-one states, but theorem 7.2 does not exclude a different route becoming cheaper under perturbation. To identify a change of mechanism, one must certify the competing upper paths and every relevant part of a common separating boundary over a parameter interval, and then prove the crossing of their exact critical heights. Two overlapping interval enclosures do not prove such a crossing.

The most direct quantitative improvement is a sharper far-field source estimate. The cubic bound $\|D\theta - \sin D\theta\|_1 \le \|\theta\|_{\mathcal{X}}^3/6$ uses a deliberately loose global energy-norm bound. Verified core sums and cancellation or multipolar information could reduce the tail constant without changing the path-space theorem. Existing far-field expansions and higher-order boundary conditions [4, 6, 17] suggest the right analytic tools; compatibility with an indefinite saddle and the exact global boundary remains an obligation, rather than a consequence of a stable-minimum approximation theorem.

The localization estimate has the form $C_E/R$, so an error allocation $\varepsilon/4$ requires $R \ge 4C_E/\varepsilon$. The current $C_E \simeq 1.60 \times 10^{17}$ explains the enormous radii: their dependence on inverse accuracy is linear. Choosing dyadic radii does not make it exponential. The separate exponential obstacle is a complete cubical covering in the number of retained variables, which is of order $R^2$. The first quantitative task is therefore to certify the actual core sum of $|q_e|$, $q = D\theta - \sin(D\theta)$, and its exterior remainder, replacing the coarse $M^3/6$ majorant. A further route is to extract leading lattice multipoles before taking absolute values. For example, with $M_j^{(q)} = \sum_y q_j(y)$ the exact identity

$$(\mathsf{P}q)_i(x) = \sum_j \mathsf{P}_{ij}(x) M_j^{(q)} + \sum_{j,y} \big[\mathsf{P}_{ij}(x-y) - \mathsf{P}_{ij}(x)\big] q_j(y)$$

separates the leading term. Improving the remainder requires explicit kernel-difference bounds, controlled source moments and exterior tails, and uniform estimates along the complete upper construction. Zero total vortex charge does not by itself establish the needed cancellations of this nonlinear source. In a perturbed problem these controls must also hold in the same admitted coupling class. We have not proved a multiphase-averaging shortcut: averaging phase representatives need not preserve their charges or the certified energy ceiling. A materializable method should retain an analytic exterior and the low-rank atlas, and would still require a verified computational budget after these sharper estimates are obtained.

## Data and software availability

The source repository is https://github.com/rsumoza/global_annihilation_barriers_infinite_-XY_lattice. The supplementary index identifies the three frozen global-verification archives by byte count and SHA-256; they are necessary in addition to the smaller source and checker snapshot. The planned frozen distribution is the tagged release `siads-v7-2026-10-22` at https://github.com/rsumoza/global_annihilation_barriers_infinite_-XY_lattice/releases/tag/siads-v7-2026-10-22, with sources in Git and the original archives attached as release assets. The repository address has been supplied by the author; publication of the tagged release and remote access remain to be verified. No DOI is asserted. The tag identifies the source commit once published; its full SHA is to be recorded in the release description and submission record. The supplement and index accompany the initial submission. Any practical portal restriction on the frozen material must be resolved through an accepted submission route; external access does not automatically waive the journal's review requirements.

## Declarations

OpenAI ChatGPT/Codex tools were used substantively in mathematical exploration and proof drafting, code development and execution, literature retrieval and comparison, adversarial checking, and manuscript preparation. The conceptual geometric figure is generated by explicit Python code; it is not a numerical output of the certificate. These automated checks and drafting assistance do not constitute independent external peer review or proof-assistant verification. The authors assume responsibility for all content.

# A Details of the explicit lattice Green bound

We justify the Fourier calculation in Lemma 5.1 with the canonical bond-anchor convention. Orthogonality of a field $f \in \mathcal{H}$ to $Dv$ for every compact $v$ is equivalent, after Fourier transform, to $\sum_i \overline{d_i}\widehat{f_i} = 0$ almost everywhere: trigonometric polynomials are dense and the $d_i$ are bounded. Thus $D(\mathcal{X})$ is the pointwise span of $d = (d_1, d_2)$, apart from the irrelevant frequency zero, and the projection symbol is

$$m_{ij}(k) = \frac{d_i(k)\overline{d_j(k)}}{\Lambda(k)}, \qquad \Lambda(k) = 4 - 2\cos k_1 - 2\cos k_2.$$

This argument does not require an $\ell^2$ vertex potential for each finite-energy gradient. In particular, $\|\mathsf{P}\|_{\mathcal{H}\to\mathcal{H}} = 1$ and every scalar symbol is bounded by one.

Let $z = (n_1, n_2)$ and first assume $r = |n_2| = \|z\|_\infty \geq 1$. For $t = k_1 \neq 0$, set

$$s = |\sin(t/2)|, \quad \alpha = 2\,\mathrm{arsinh}\, s, \quad g_n(t) = \frac{e^{-\alpha|n|}}{2\sinh\alpha}.$$

Since $\cosh\alpha = 2 - \cos t$, the Poisson kernel identity gives

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{e^{iny}}{4 - 2\cos t - 2\cos y}\, dy = g_n(t). \tag{78}$$

It follows by expanding $(\cosh\alpha - \cos y)^{-1}$ as a geometric Fourier series; all series converge absolutely for this fixed $t \neq 0$.

The partially integrated diagonal entries are $4s^2 g_{n_2}$ and $\delta_{n_2,0} - 4s^2 g_{n_2}$. Since $n_2 \neq 0$, their magnitudes are at most

$$4s^2 g_{n_2} = \frac{s}{\sqrt{1+s^2}} e^{-\alpha r} \leq s e^{-\alpha r}.$$

For the 12 entry, the partial coefficient is $d_1[g_{n_2-1} - g_{n_2}]$; for the 21 entry it is $\overline{d_1}[g_{n_2+1} - g_{n_2}]$, up to the choice of Fourier sign. For both shifts, including $r = 1$,

$$|g_{n_2 \pm 1} - g_{n_2}| \leq \frac{e^{\alpha} - 1}{2 \sinh \alpha} e^{-\alpha r} = \frac{e^{\alpha}}{e^{\alpha} + 1} e^{-\alpha r} \leq e^{-\alpha r}. \tag{79}$$

As $|d_1| = 2s$, the two mixed entries have partial magnitude at most $2se^{-\alpha r}$. Thus no unit shift of the anchor distance remains unaccounted for.

For $0 \leq t \leq \pi$, $s \leq t/2$, $s \geq t/\pi$, and $\operatorname{arsinh} s \geq s/\sqrt{2}$. The last inequality follows by integrating $(1+s^2)^{-1/2} \geq 1/\sqrt{2}$ on $[0, s] \subset [0, 1]$. Consequently

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} s e^{-\alpha r} \, dt \leq \frac{1}{2\pi} \int_0^{\infty} t e^{-\sqrt{2} r t/\pi} \, dt = \frac{\pi}{4r^2}.$$

This gives $|\mathsf{P}_{ij}(z)| \leq \pi/(2r^2) < 2/r^2$. If $|n_1| = \|z\|_{\infty} \geq 1$, exchange the axes and their orientation labels. At $z = 0$ use the integral of the bounded scalar symbol. The multiplier entries are bounded on a finite measure torus, so Fubini applies directly. The exceptional value $t = 0$ has measure zero, and the partial bounds above are integrable. There is no appeal to an uncontrolled logarithmically divergent vertex Green function.

# B An annular Poincaré constant for the cutoff

For $R \geq 1$ integer and $\mathcal{A}_R = \{x : R \leq \|x\|_{\infty} \leq 2R\}$, we prove

$$\sum_{\mathcal{A}_R} |u - \overline{u}_{\mathcal{A}_R}|^2 \leq \frac{225}{4} R^2 \sum_{e \subset \mathcal{A}_R} |D_e u|^2. \tag{80}$$

For $n$ values on an interval, the variance identity and Cauchy–Schwarz give

$$\sum_i |u_i - \overline{u}|^2 = \frac{1}{n} \sum_{i<j} |u_i - u_j|^2 \leq \sum_{k=1}^{n-1} \frac{k(n-k)}{2} |u_{k+1} - u_k|^2 \leq \frac{n^2}{8} \sum_k |Du_k|^2.$$

Decomposing the variance of an $n_1 \times n_2$ rectangle into row variances and the variance of row means, followed by Jensen's inequality, gives the constant $\max(n_1^2, n_2^2)/8$ for its full edge energy.

Cover the annulus by its four top, right, bottom, and left strips, each with $(4R+1)(R+1) = N$ vertices. Adjacent strips have an overlap of $J = (R+1)^2$ vertices. Write $m_i$ for their means and $V_i = \sum_{S_i} |u - m_i|^2$. Averaging over an overlap yields $|m_i - m_j|^2 \leq 2(V_i + V_j)/J$. If $c$ is the mean of the four $m_i$, the four-cycle inequality gives

$$\sum_i |m_i - c|^2 \leq \frac{1}{2} \sum_{i \sim j} |m_i - m_j|^2 \leq \frac{2}{J} \sum_i V_i.$$

Hence

$$\sum_{\mathcal{A}_R} |u - c|^2 \leq \sum_i V_i + N \sum_i |m_i - c|^2 \leq (1 + 2N/J) \sum_i V_i \leq 9 \sum_i V_i.$$

Each edge belongs to at most two strips and $(4R+1)^2 \leq 25R^2$. The rectangular estimate bounds the last expression by $9 \cdot 2 \cdot 25R^2/8$ times the annular edge energy. Replacing $c$ by the annular mean can only decrease the variance, proving (80).

# C Explicit dipolar and payload norm bounds

Let $L = 12$ and let $\phi_{12}$ be the real reference in section 5. For an index radius $r \geq 16$, the distance to every interpolated pole is at least $r - 8 \geq r/2$. Integrating the derivative of the argument with respect to pole position, and then along a unit bond, gives

$$|\phi_{12}(x)| \leq 2L/r, \qquad |D_e\phi_{12}| \leq 4L/r^2 \quad \text{if } \min(\|x\|_\infty, \|y\|_\infty) = r.$$

These integrations take place away from the pole segment, so they do not introduce branch jumps. There are at most $16r + 4 \leq 20r$ bonds with minimum endpoint radius $r$. Therefore, for integer $R \geq 16$,

$$\sum_{\min(\|x\|_\infty, \|y\|_\infty) \geq R} |D_e\phi_{12}|^2 \leq 320L^2 \sum_{r \geq R} r^{-3} \leq 480L^2/R^2.$$

For the cutoff $\zeta_R$ used in lemma 5.4, the annulus has $12R(R+1) \leq 24R^2$ vertices. Thus direct use of its product identity gives

$$\|(1 - \zeta_R)\phi_{12}\|_{\mathcal{X}}^2 \leq 2(480L^2/R^2) + \frac{4}{R^2} \sum_{\mathcal{A}_R} |\phi_{12}|^2 \leq 1344L^2/R^2.$$

Since $\sqrt{1344} < 40$, best approximation proves $\|(I - \Pi_{2R})\phi_{12}\|_{\mathcal{X}} \leq 40L/R = 480/R$.

For completeness, a coarse global bound is

$$\|\phi_{12}\|_{\mathcal{X}}^2 \leq 4 \cdot 33^2(4\pi)^2 + 480 \cdot 12^2/16^2 < 10^6.$$

Here the first term overcounts every bond touching $\Lambda_{16}$; each difference of the two real arguments is at most $4\pi$ in magnitude. For a compact correction on an $n \times n$ vertex array,

$$\|h\|_{\mathcal{X}} \leq \sqrt{8}\, \|h\|_{\ell^2} \leq 3n \max |h|.$$

Exact binary-payload inspection, including finiteness, zero boundary values, and manifest hashes, gives

$$\max |h_m| = \frac{99633080638867}{562949953421312} < .177,$$

$$\max |h_s| = \frac{1834512834580789}{4503599627370496} < .4074,$$

$$\max_j \max |z_j| = \frac{6674417597839975}{2251799813685248} < 3.$$

The minimum and saddle arrays have side 2048 and 4096, respectively; the stored path displacements $z_j$ have side 256 and are added to the minimum candidate. Hence

$$\|m_0\|_{\mathcal{X}} < 1000 + 3(2048)(.177) + .017422 < 6007,$$

$$\|s_+\|_{\mathcal{X}} < 1000 + 3(4096)(.4074) + .012825 < 6007,$$

$$\max_j \|\phi_{12} + h_m + z_j\|_{\mathcal{X}} < 1000 + 3(2048)(.177) + 3(256)(3) < 6007.$$

Reflection preserves the energy norm. Linearity and the convexity of the norm cover all segments, the two exact attachments, and the final contraction to zero. The original two candidates coincide with $\phi_{12}$ beyond minimum-endpoint radius 4096, where its increments are smaller than $10^{-6}$. Adding the certified correction radii gives the common bound .018 used in section 5. The supplied payload checker reproduces the array bounds from the two frozen archives; the stationary correction radii still depend on the separate full-space certificate.

# D The full dual residual of the neutral reference

This calculation justifies the exterior residual formula used in the fixed critical-state certificate. In centered coordinates $(\xi, y) = (x_1 - 6, x_2)$, set $a = L/2 = 6$ and

$$\varphi(\xi, y) = \arg(\xi + a + \mathrm{i}y) - \arg(\xi - a + \mathrm{i}y), \qquad L = 12.$$

Use the real branch tending to zero away from the finite cut $[-a, a] \times \{0\}$. Its gradient is square summable; compact cutoffs converge in the energy norm since $\varphi = O(L/r)$ and $D\varphi = O(L/r^2)$ outside that cut. Thus $\varphi \in \mathcal{X}$. Each unoriented bond is counted once, with $D_e u = u(e^+) - u(e^-)$, and $D^*$ denotes the vertex-distribution adjoint. Write $f_\theta = D^* \sin(D\theta)$.

Let $S > a + 2$ be an integer and define the symmetric vertex square

$$B_S = \{v \in (\mathbb{Z} + 1/2)^2 : \|v\|_\infty \le S - 3/2\}, \qquad b = 2S - 2, \qquad \chi = \frac{S}{S - a - 2}.$$

Here the square is in the centered coordinates. Suppose $h$ is compact and every bond on which $D_e h \neq 0$ has *both* endpoints in $B_S$. Set $f_B = \mathbf{1}_{B_S} f_{\varphi+h}$, interpreted as a load on the full lattice. Then $f_B$ has algebraically zero mean. Indeed, $j_e = \sin D_e(\varphi + h) - \sin D_e\varphi$ is a compact flux whose nonzero bonds and both endpoints lie in $B_S$, so $\sum_{B_S} D^* j = 0$ by endpoint cancellation. The pure-reference phase is complex-conjugated by $(\xi, y) \mapsto (\xi, -y)$; its vertex force is therefore odd under this reflection. Pairing the vertices of $B_S$ gives $\sum_{B_S} f_\varphi = 0$. No oddness assumption on $h$ is needed. An interval sum containing zero would not replace this argument.

Let $L_N$ and $D_N$ count only bonds interior to $B_S$. For any finite proposal $u$, the Neumann comparison and $\lambda_1(L_N) = 4\sin^2(\pi/(2b)) \ge 4/b^2$ give

$$\|f_B\|_{\mathcal{X}^*} \le \|D_N u\|_2 + \frac{b}{2} \|f_B - L_N u\|_2 . \tag{81}$$

To see this, solve $L_N u_* = f_B$ with the same mean as $u$, and use $\|f_B\|_{\mathcal{X}^*} \le \|D_N u_*\|_2$ and $\|D_N(u_* - u)\|_2 \le \lambda_1(L_N)^{-1/2} \|f_B - L_N u\|_2$. No normalization of the mean of $u$ is required.

The complementary load obeys the explicit bound

$$\|f_{\varphi+h} - f_B\|_{\mathcal{X}^*} \le \frac{64\sqrt{3}}{5} \frac{L\chi^5}{(S-1)^{5/2}} + \frac{32\sqrt{3}}{9} \frac{L^3\chi^7}{(S-1)^{9/2}}. \tag{82}$$

Here is a derivation including the cancellation needed for the second term. In a differentiation region at distance at least $d$ from the cut, subtracting the complex derivatives of the two argument fields gives

$$|\partial_i^r \varphi| \le r! L/d^{r+1} \quad (r \ge 1), \qquad \Delta\varphi = 0.$$

For $p_i = \varphi(v + e_i) - \varphi(v)$ and $m_i = \varphi(v) - \varphi(v - e_i)$, the central Taylor remainder of order four and harmonicity yield $|L_\infty \varphi(v)| \le 4L/d^5$. Also $|p_i|, |m_i| \le L/d^2$ and $|p_i - m_i| \le 2L/d^3$. Since $g(t) = \sin t - t$ has $|g'(t)| \le t^2/2$,

$$|g(p_i) - g(m_i)| \le L^3/d^7, \qquad |f_\varphi(v)| \le 4L/d^5 + 2L^3/d^7.$$

The two opposite bonds are combined before taking absolute values.

Exterior shells have $\|v\|_\infty = k - 1/2$, $k \ge S$, and at most $8k$ vertices. The unit segments used above lie at distance at least $k - a - 3/2 \ge k - a - 2$ from the cut, while $(k - a - 2)^{-r} \le \chi^r k^{-r}$. Pair each exterior vertex with $y > 0$ with its reflected vertex $Rv$. The load $\delta_v - \delta_{Rv}$ has $\mathcal{X}^*$ norm at most $\sqrt{2|y|} \le \sqrt{3k}$, by Cauchy–Schwarz along the vertical lattice path. This path may enter $B_S$;

only its endpoints specify the exterior load. Since $h$ changes no exterior vertex force, the triangle inequality gives

$$\|f_{\varphi+h} - f_B\|_{\mathcal{X}^*} \le 32\sqrt{3}L\chi^5 \sum_{k\ge S} k^{-7/2} + 16\sqrt{3}L^3\chi^7 \sum_{k\ge S} k^{-11/2}.$$

Integrating these positive decreasing terms from $S-1$ proves (82). The convergent majorant justifies the load series in $\mathcal{X}^*$; equality with the exterior vertex force is first checked on compact tests and then extended by density.

In the checker the correction array has even side $n$, its boundary is zero, the phase box has side $n+4$, and the force box has side $b = n+2$. Its halo includes every changed flux bond and both endpoints, and all four neighbours of every force-box vertex are present. Thus $S = b/2 + 1$ and $\chi = S/(S-8)$ agree exactly with the implemented formula. Adding (81) and (82) bounds the full dual residual.

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